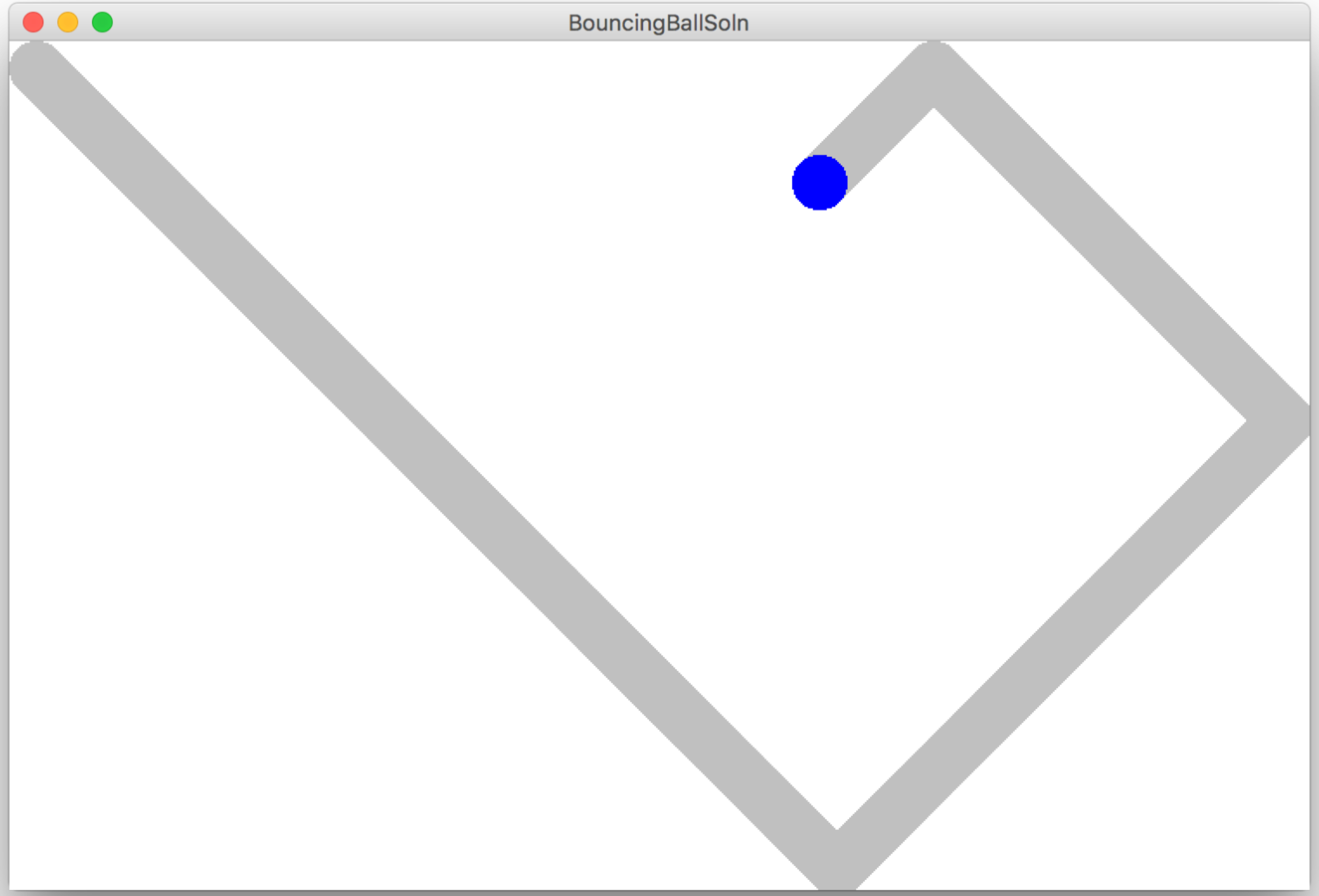
Learning Goals

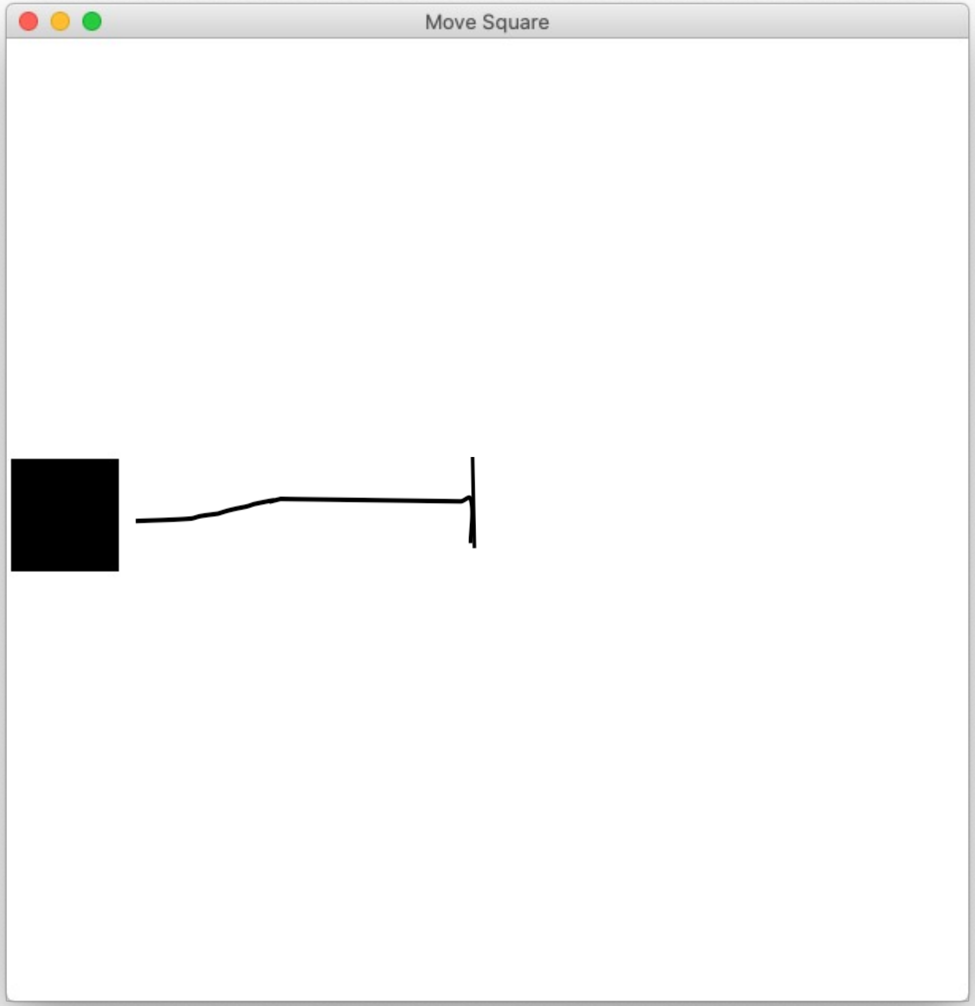
1. Write animated programs

You will be able to write Bouncing Ball



Great foundation

Move to Center



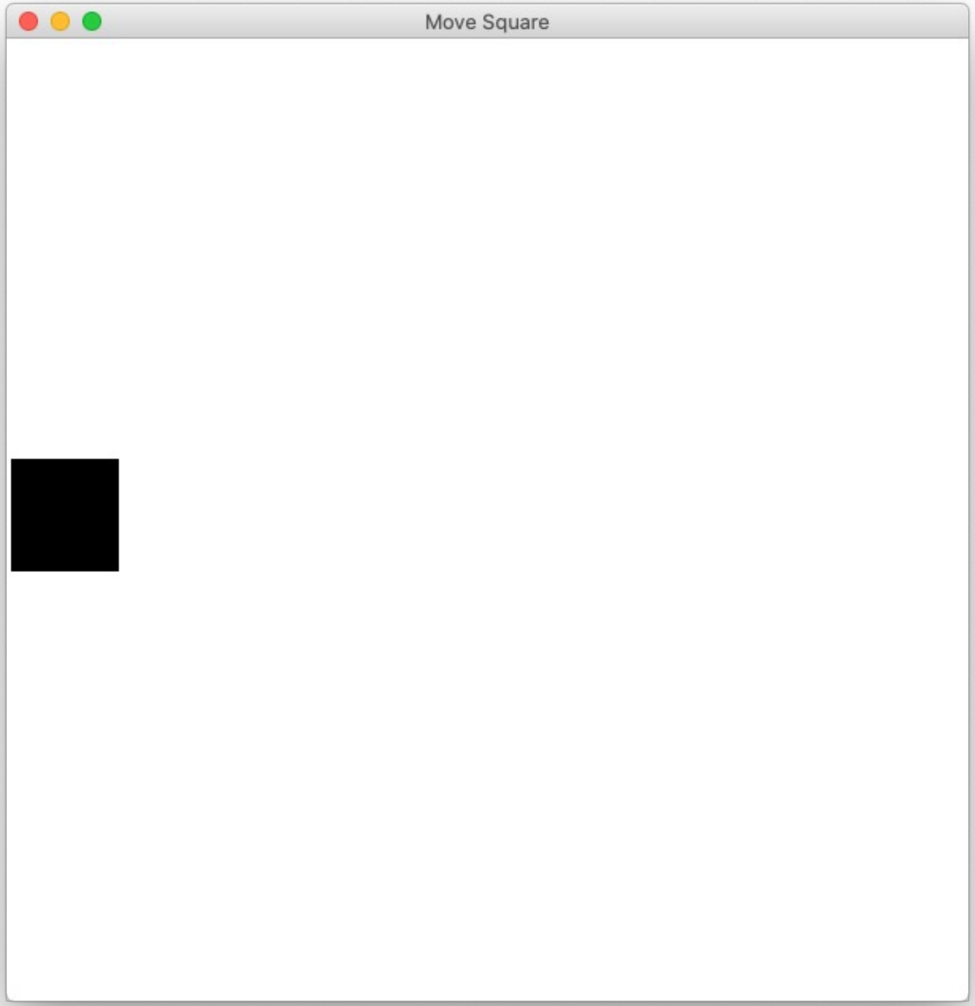
In our last episode...

Graphics

```python
from graphics import Canvas
```

```python
# we usually write this for you, and include
# it in your projects!
canvas = Canvas(width, height)
```

Add square



```python
def main():
    canvas = Canvas(CANVAS_WIDTH, CANVAS_HEIGHT)
    start_x = 0
    start_y = CANVAS_HEIGHT / 2 - SQUARE_SIZE / 2
    square = canvas.create_rectangle(start_x, start_y,
                                      SQUARE_SIZE, start_y + SQUARE_SIZE)
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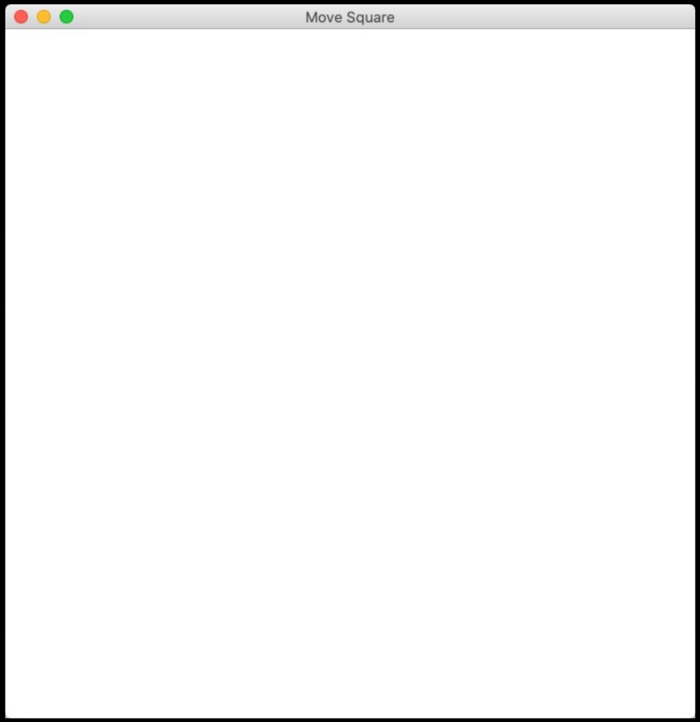

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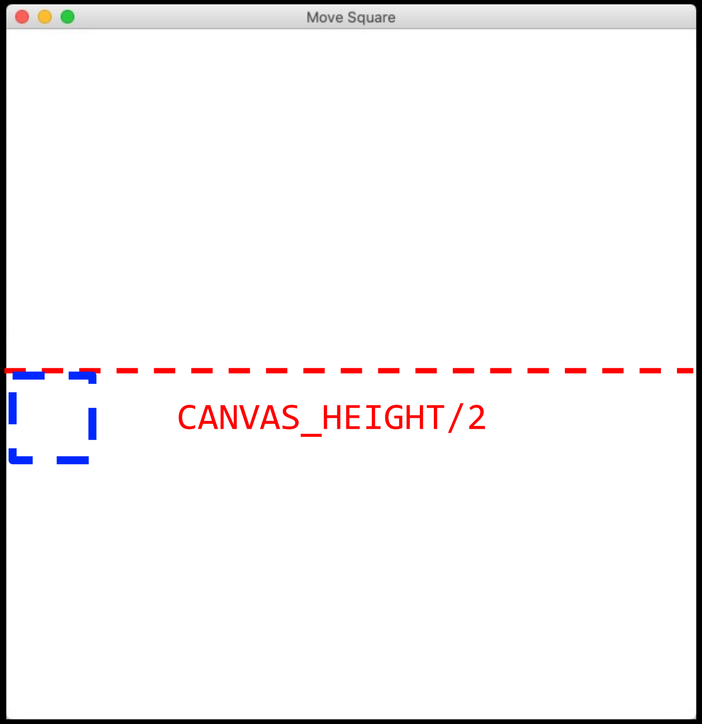

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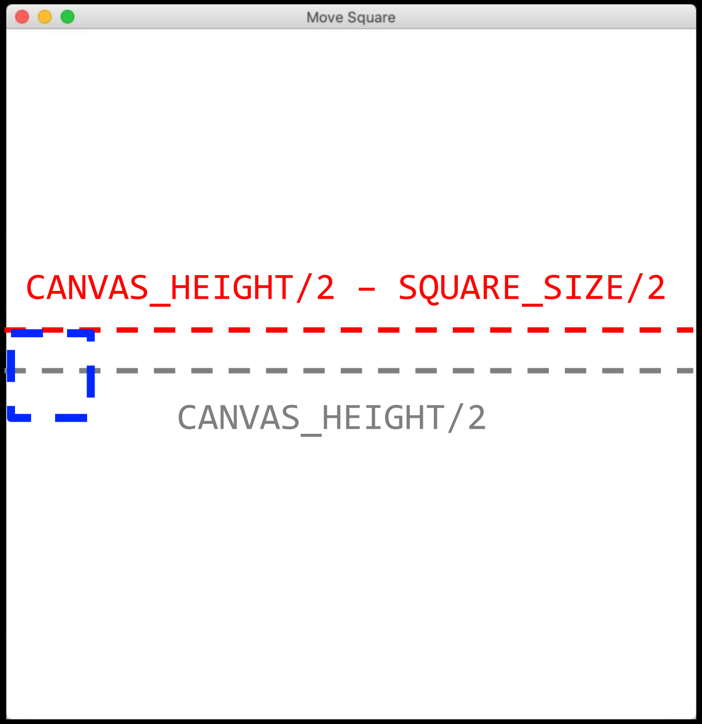

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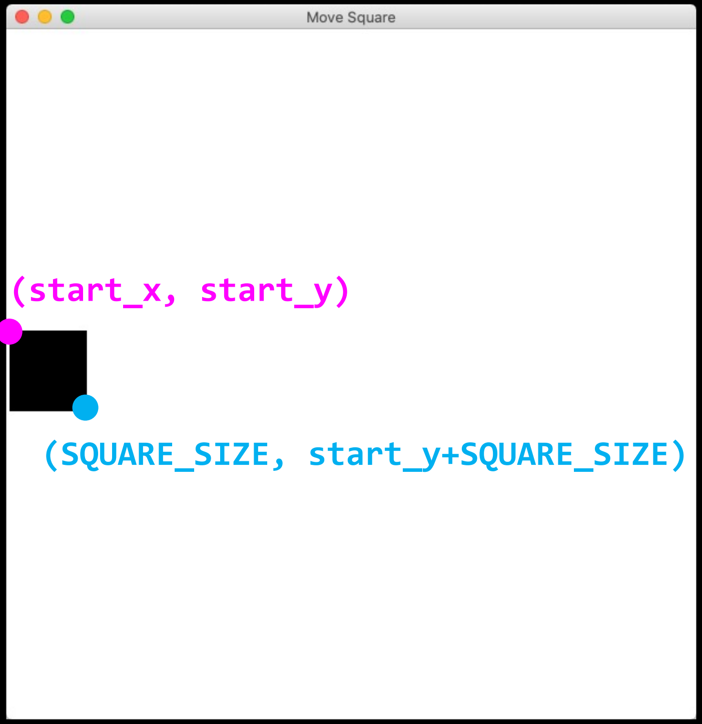
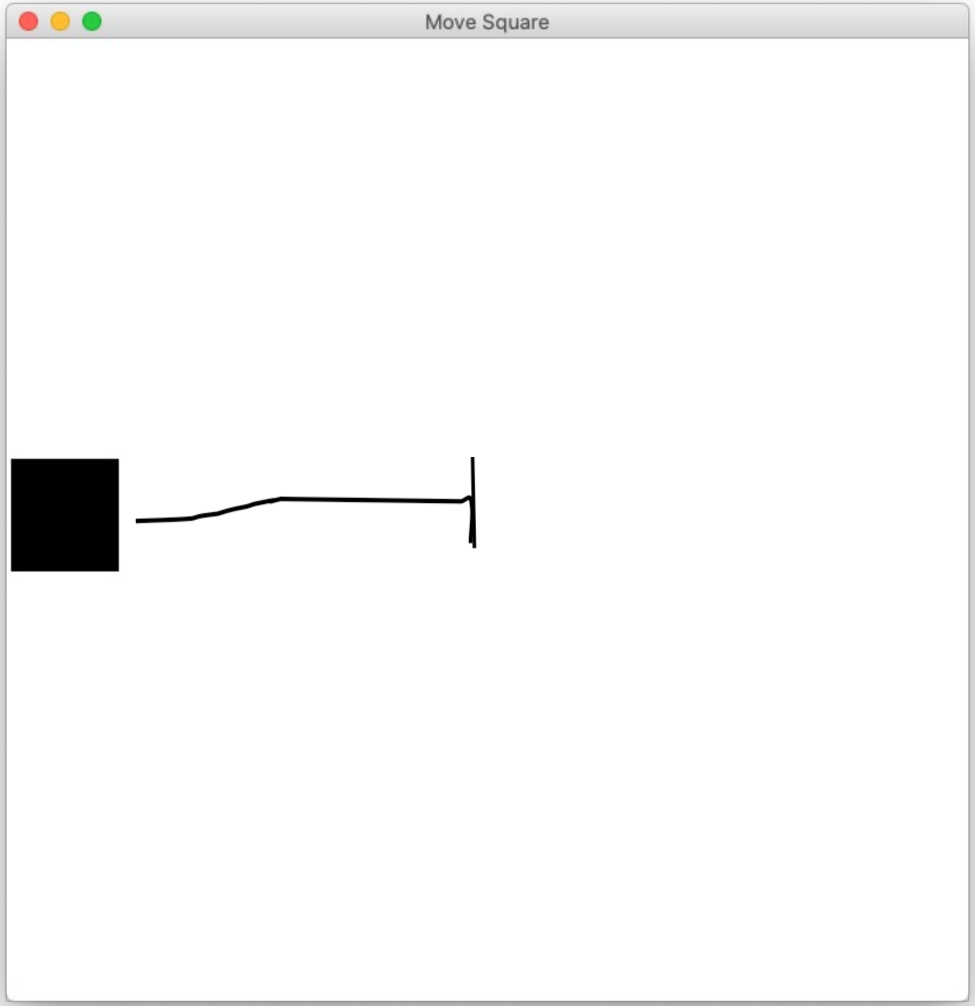


We got the square!

Um... What about the animation?!

How do movies or games animate?

Move to Center



* That's not quite Disney, but it is a start…

Animation Loop

```
def main():
    # setup

    while True:
        # update world

        # pause
        time.sleep(DELAY)
```

Animation Loop

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Make all the variables you need.

Animation Loop

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    while True:
        # update world

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        time.sleep(DELAY)
```

The animation loop is a repetition of heartbeats

Animation Loop

```
def main():
    # setup

    while True:
        # update world

        # pause
        time.sleep(DELAY)
```

Each heart-beat, update the world forward one frame

Animation Loop

```
def main():
    # setup

    while True:
        # update world

        # pause
        time.sleep(DELAY)
```

If you don't pause, humans won't be able to see it

Moving square in action!

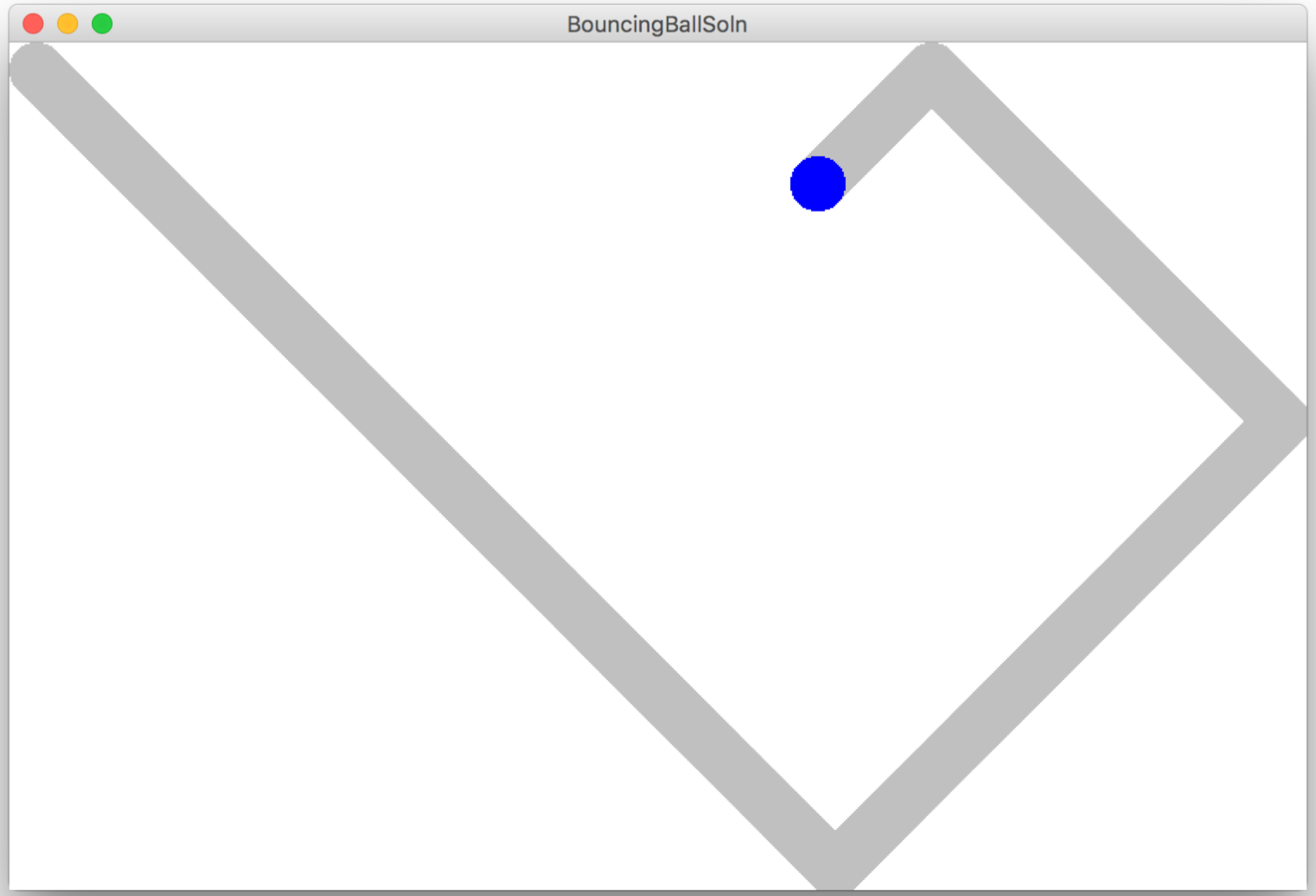
Let's see some code!

Move To Center

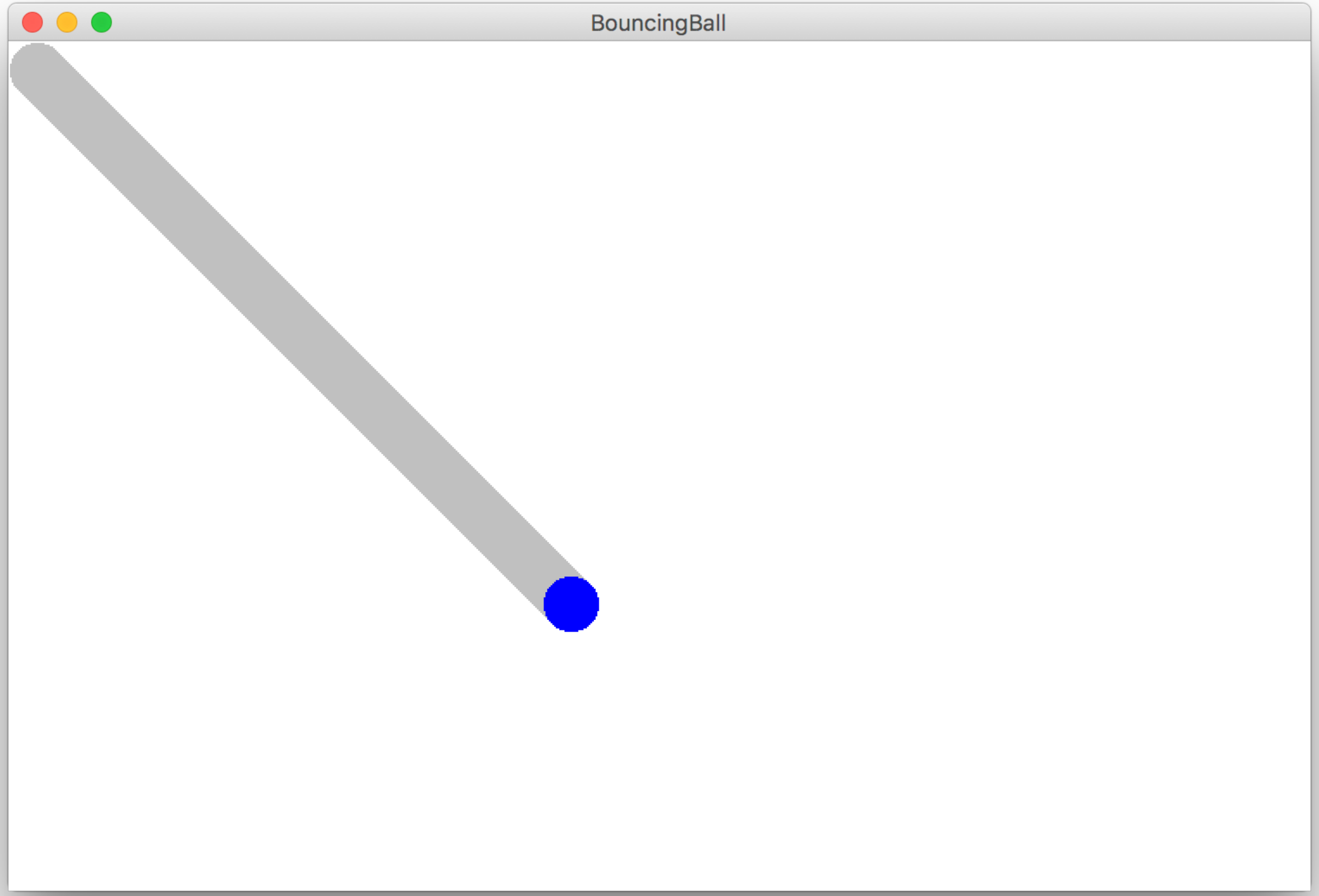
```
VELOCITY = 2

def main():
    # setup
    canvas = Canvas(CANVAS_WIDTH, CANVAS_HEIGHT)
    start_x = 0
    start_y = CANVAS_HEIGHT / 2 - SQUARE_SIZE / 2
    square = canvas.create_rectangle(start_x, start_y,
                        SQUARE_SIZE, start_y + SQUARE_SIZE)

    while (start_x + SQUARE_SIZE / 2) < CANVAS_WIDTH / 2:
        start_x += VELOCITY
        canvas.moveto(square, start_x, start_y)

        # pause
        time.sleep(DELAY)
```

We are ready…

Bouncing Ball

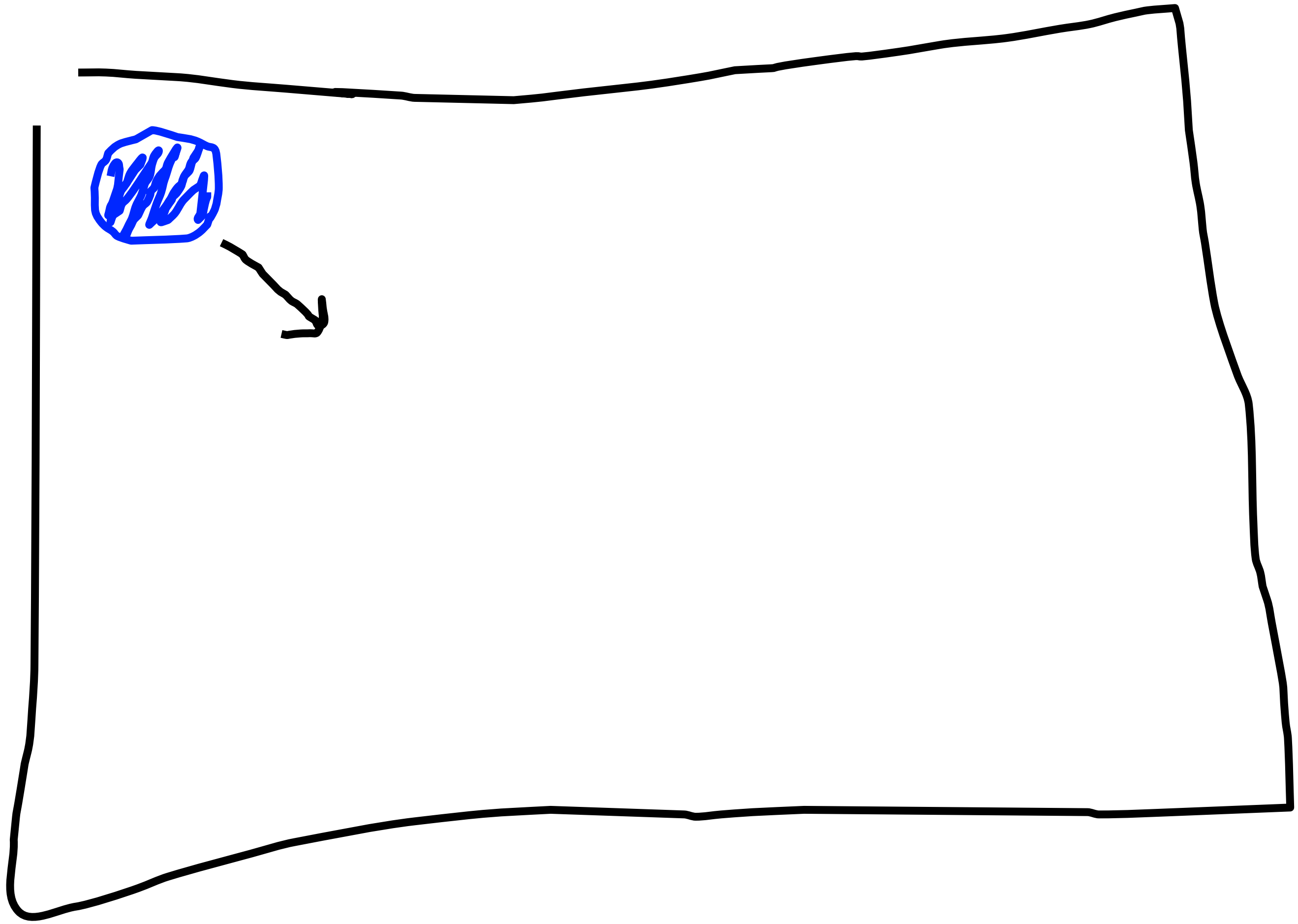


Milestone #1



Bouncing Ball

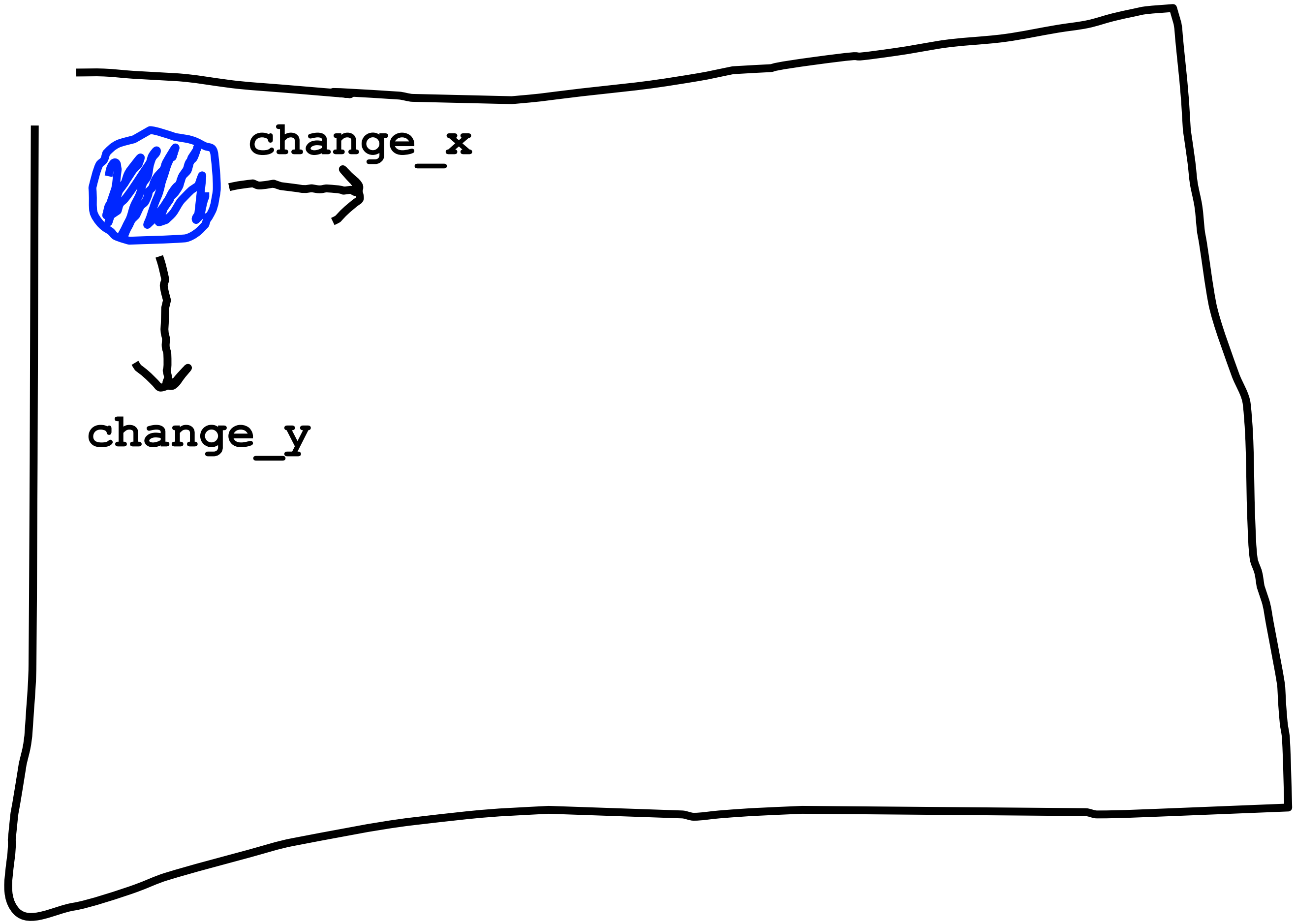
First heartbeat



Key variable: how much the ball position change each heartbeat?

Bouncing Ball

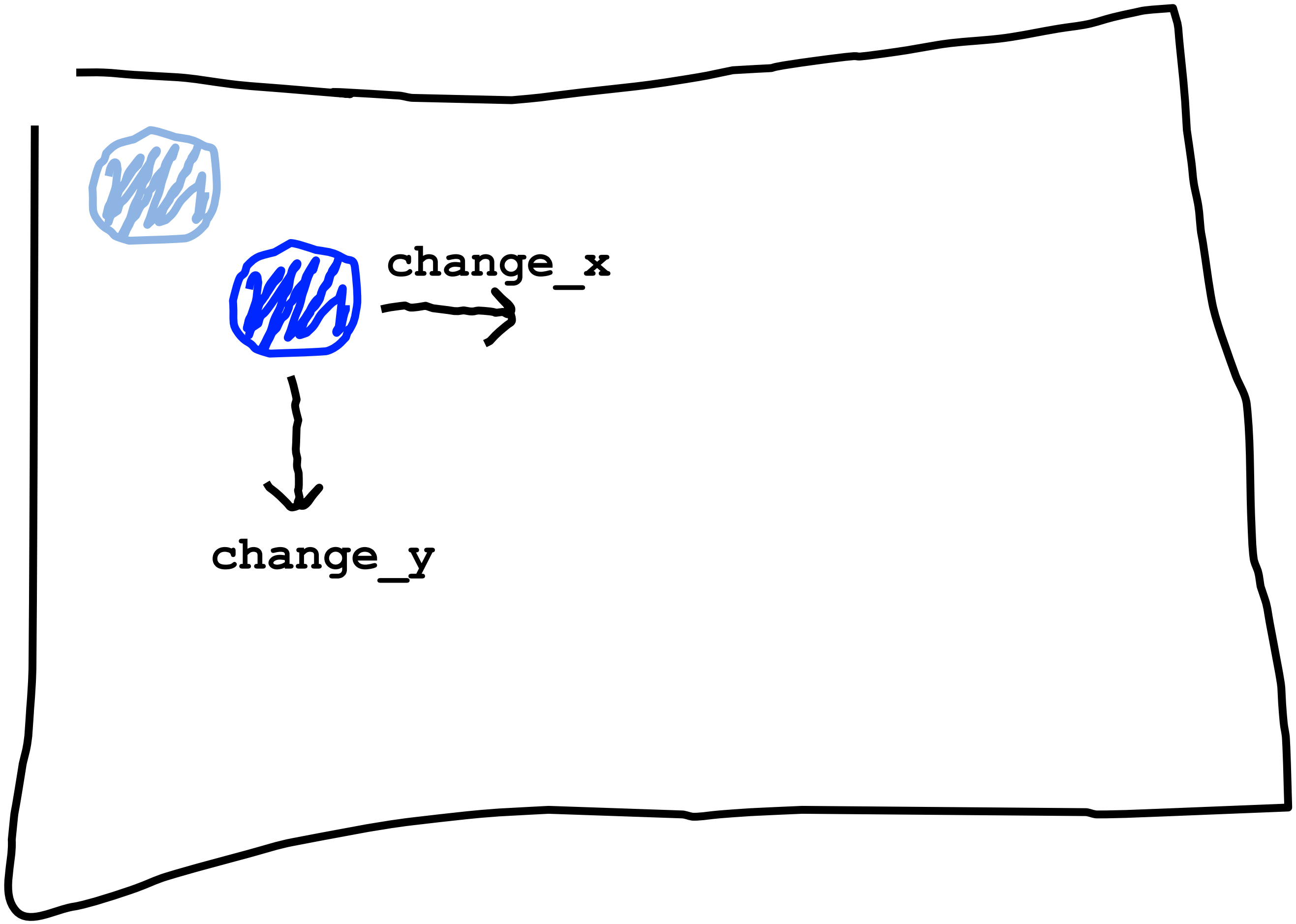
First heartbeat



The **`move`** function takes in
a change in x and a change in y

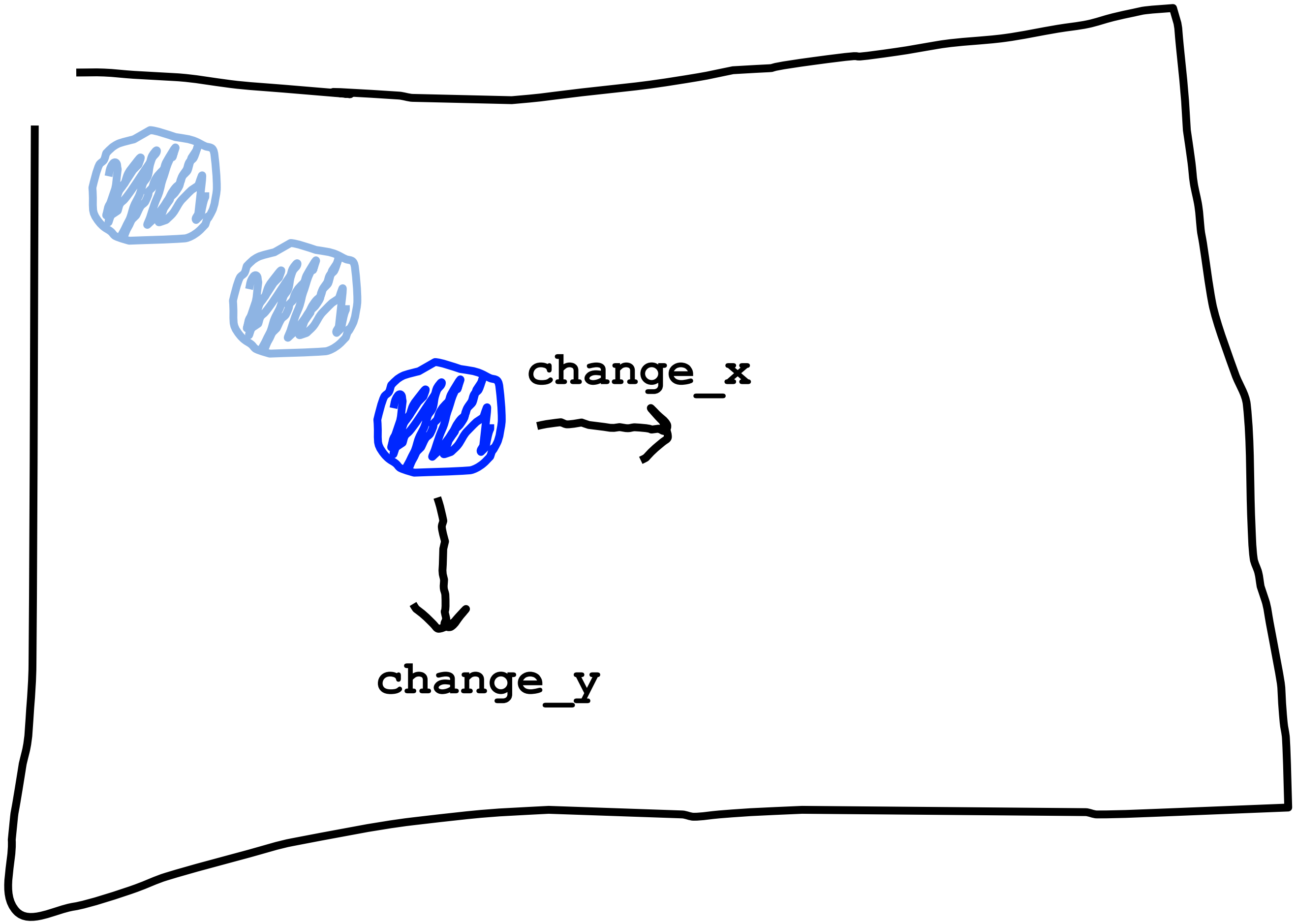
Bouncing Ball

Second heartbeat



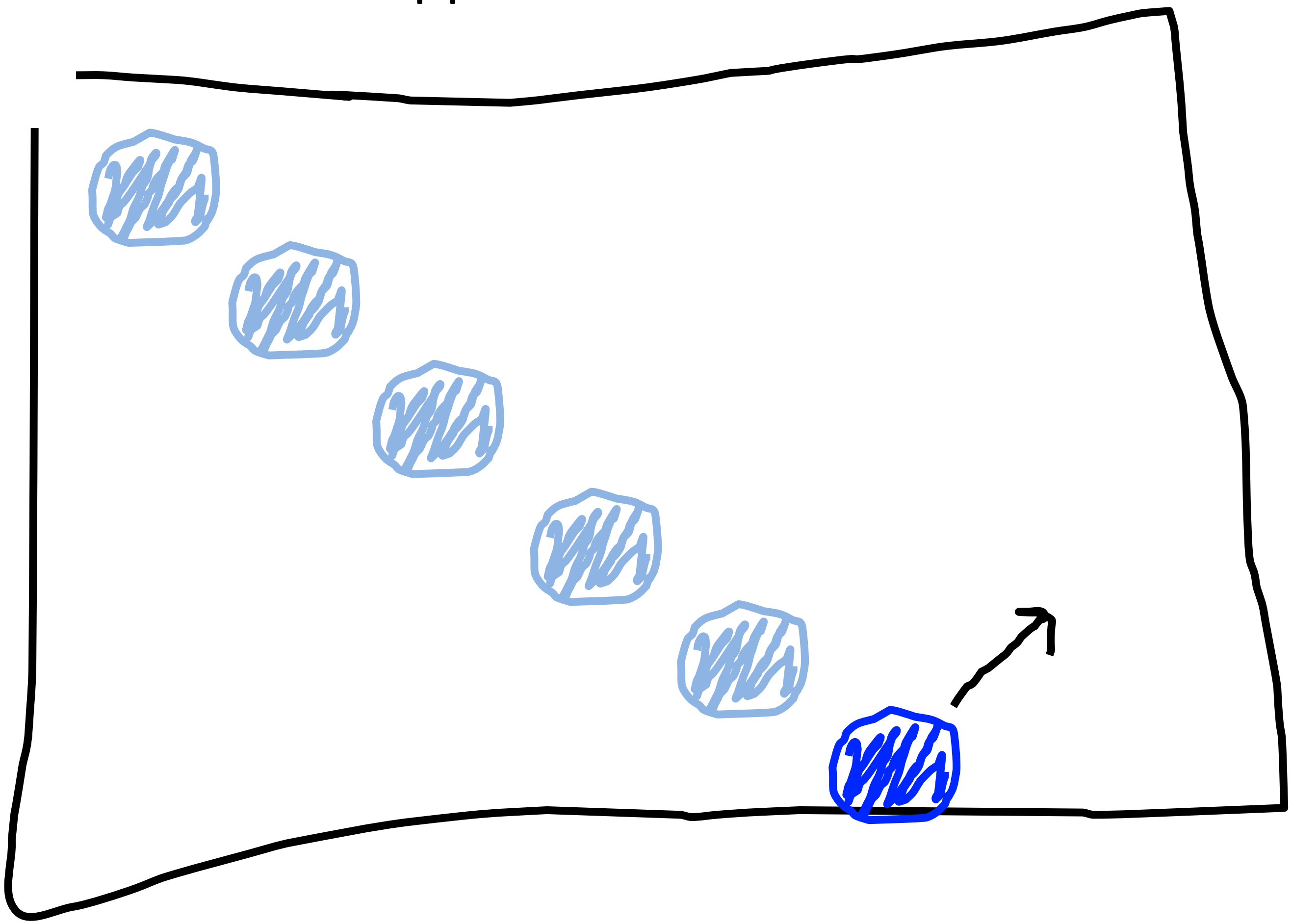
Bouncing Ball

Third heartbeat



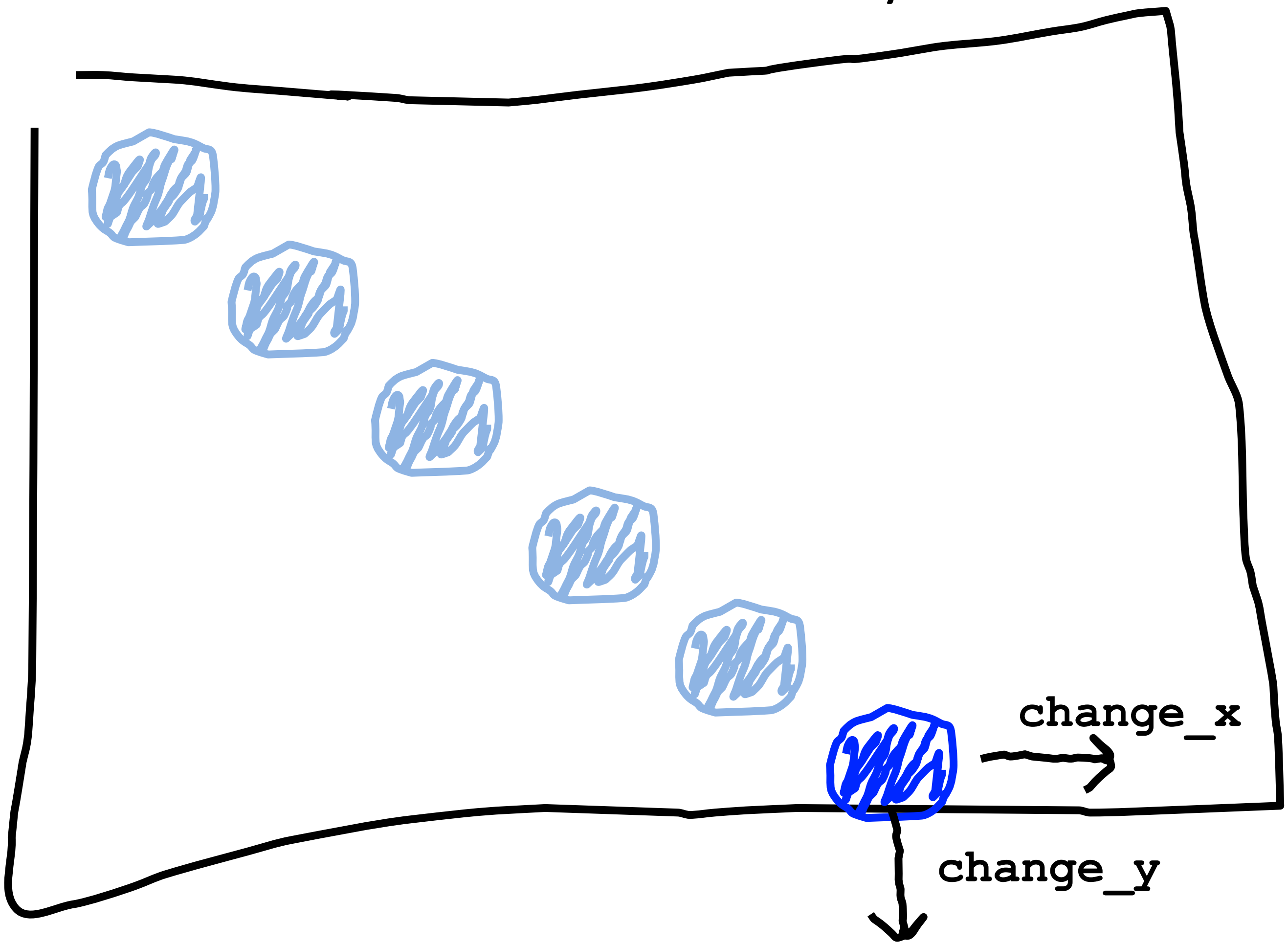
Bouncing Ball

What happens when we hit a wall?



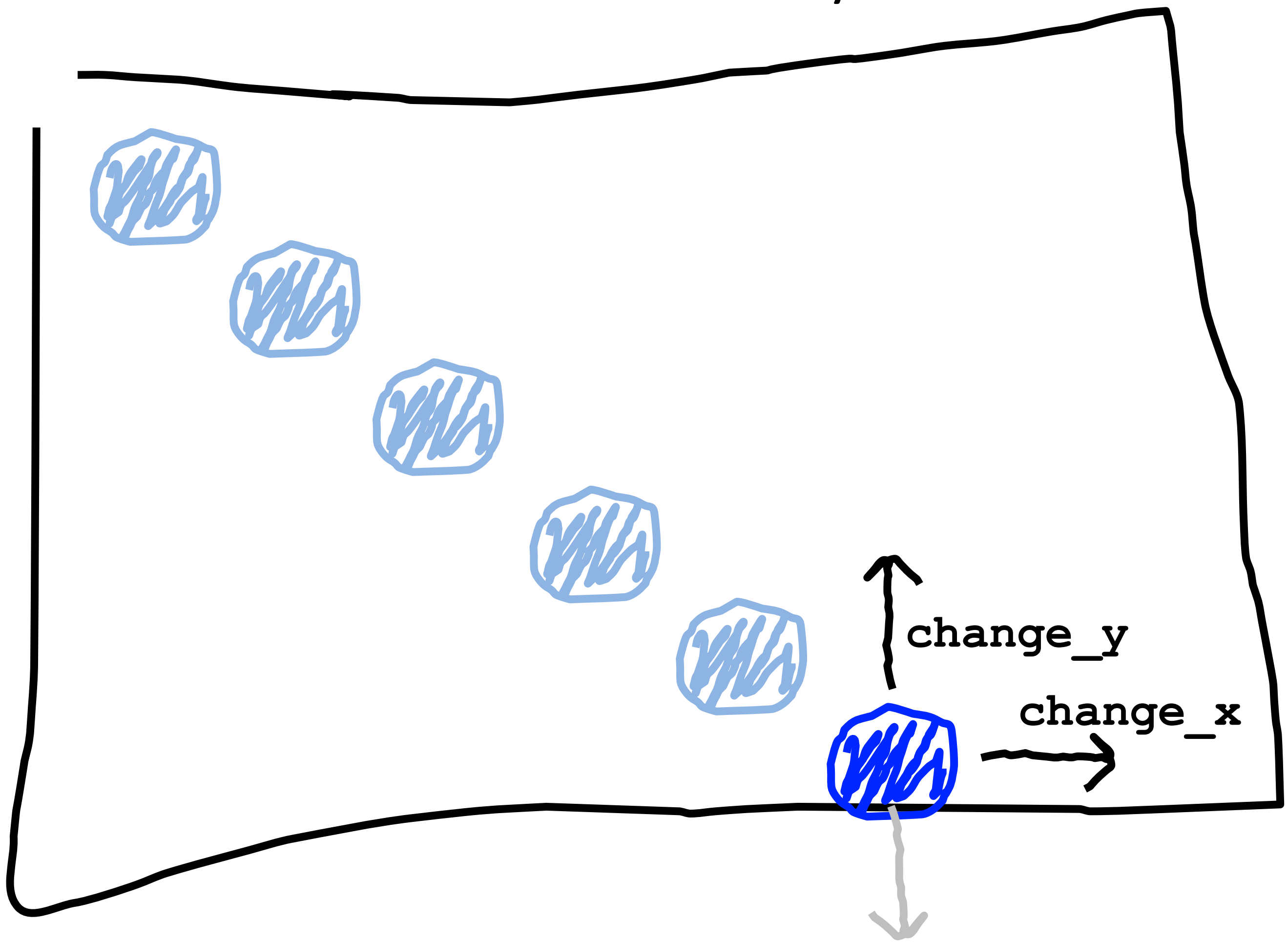
Bouncing Ball

We have this velocity



Bouncing Ball

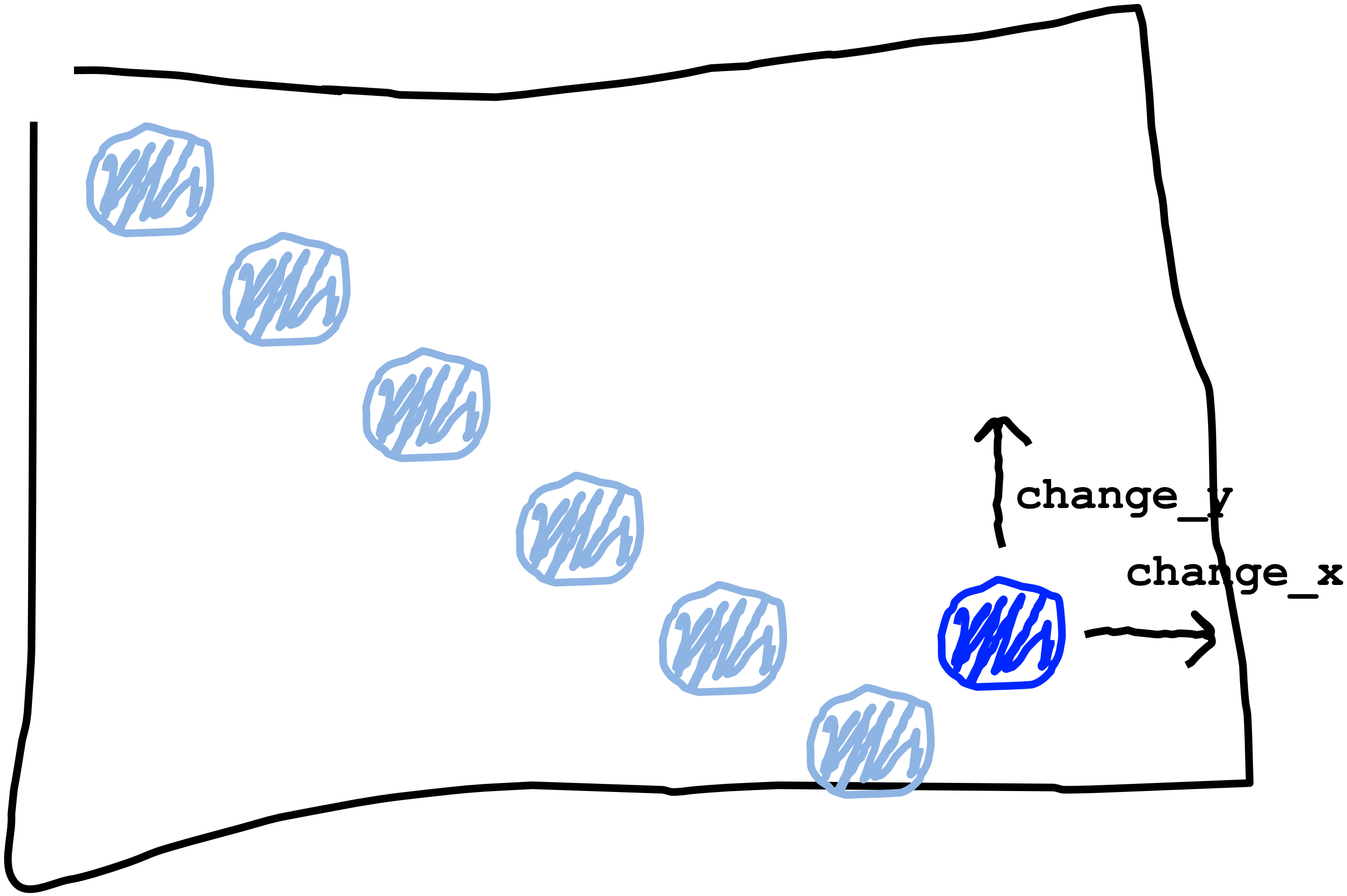
Our new velocity



When reflecting vertically: **`change_y = -change_y`**

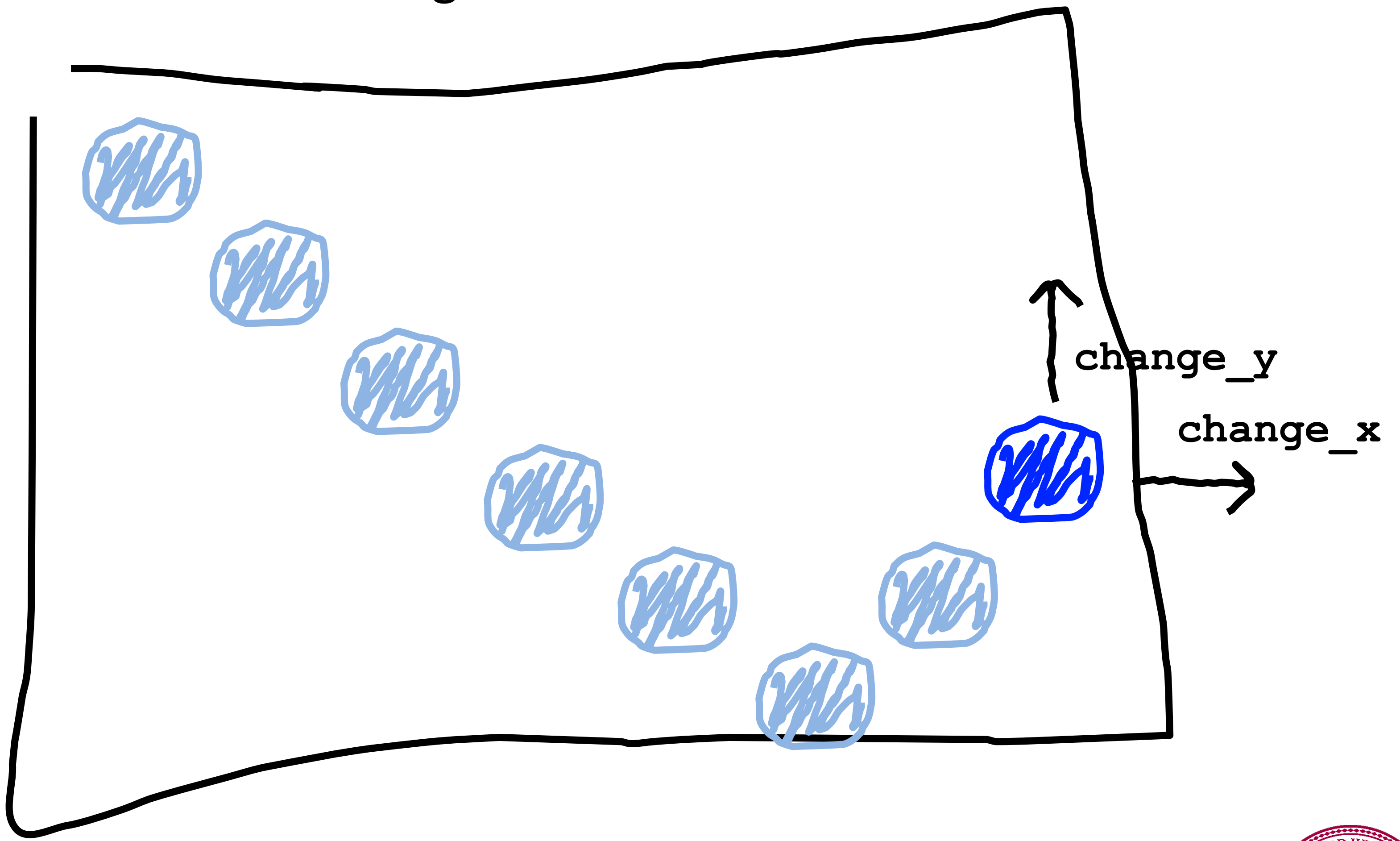
Bouncing Ball

Seventh heartbeat



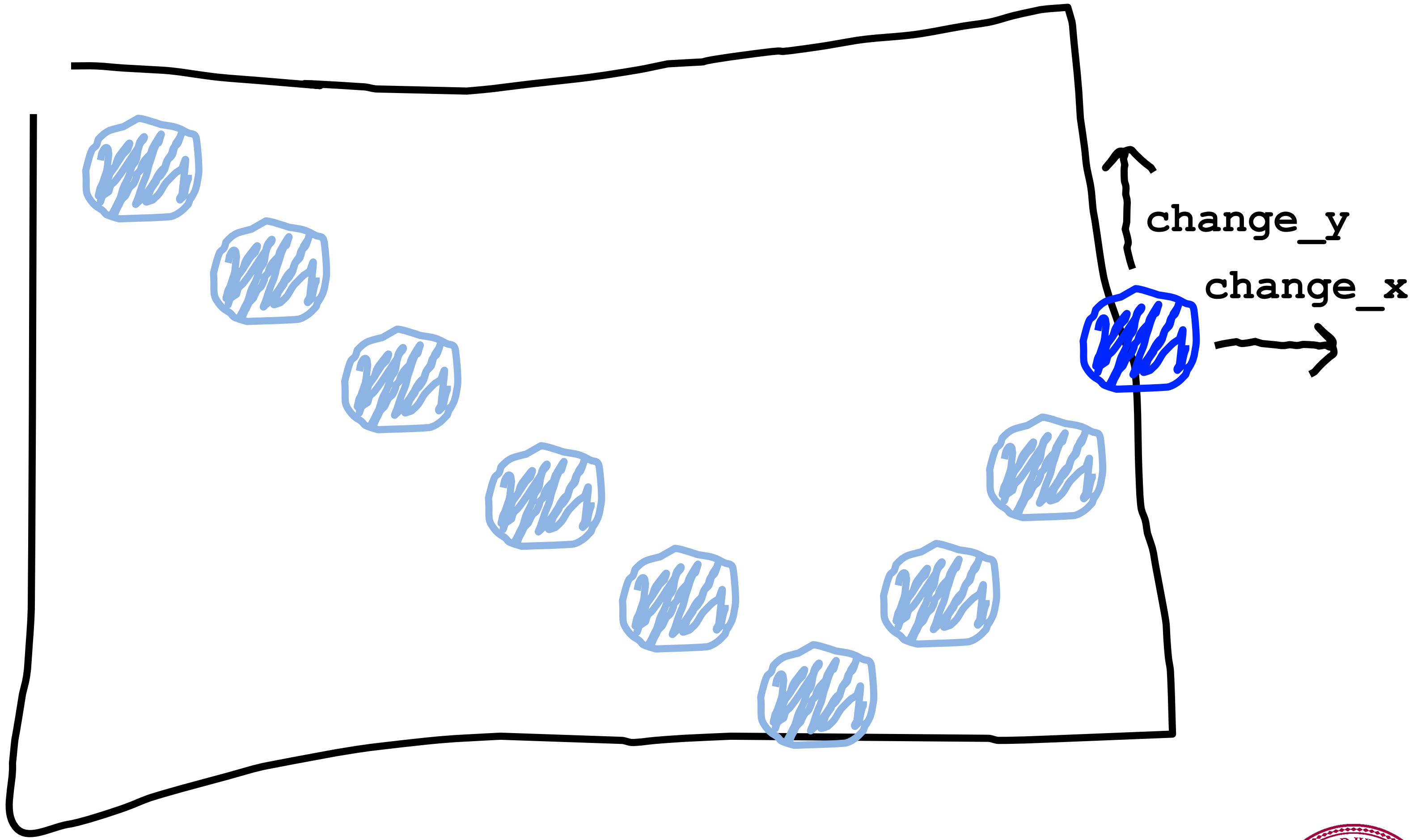
Bouncing Ball

Eighth heartbeat



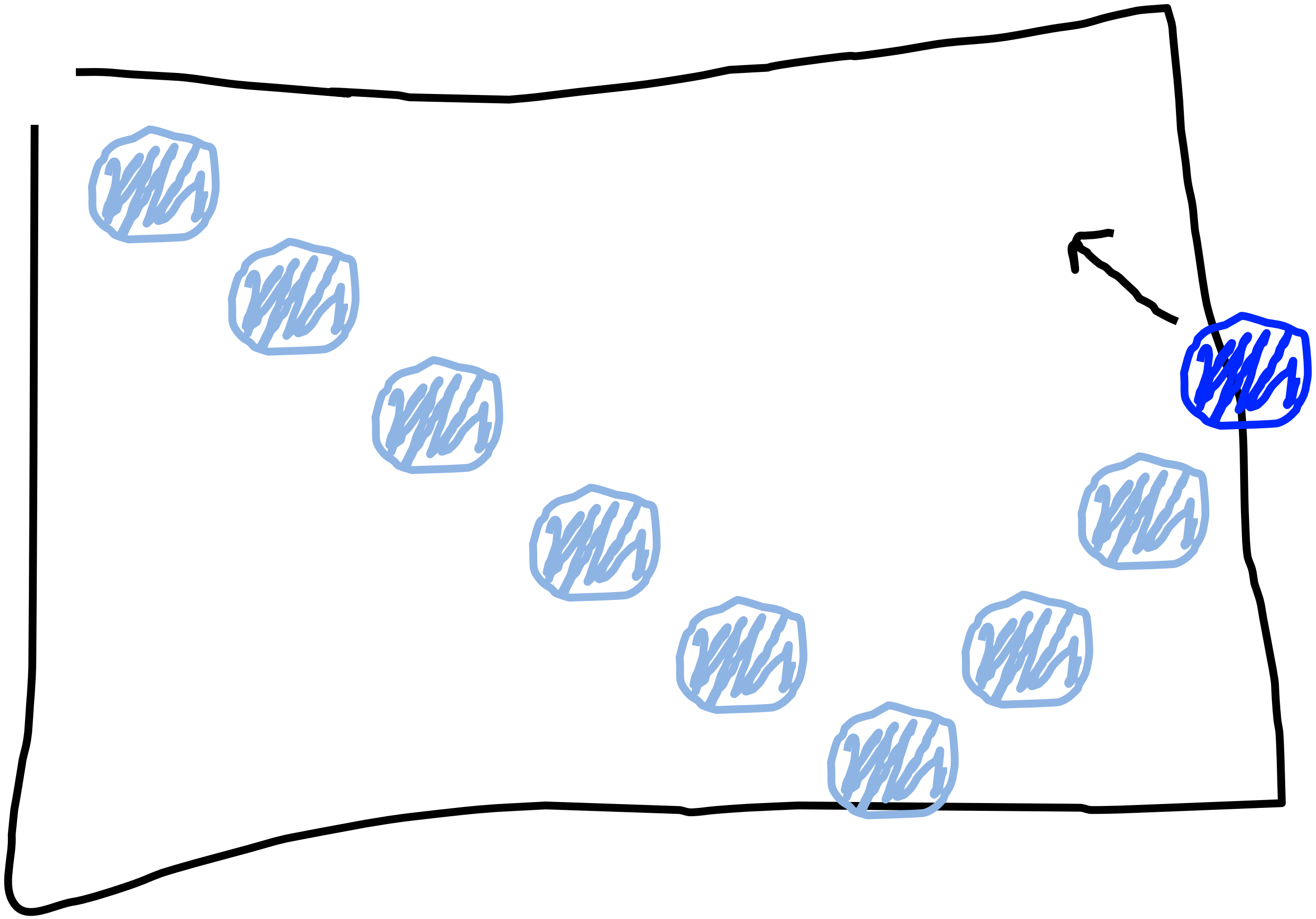
Bouncing Ball

Ninth heartbeat



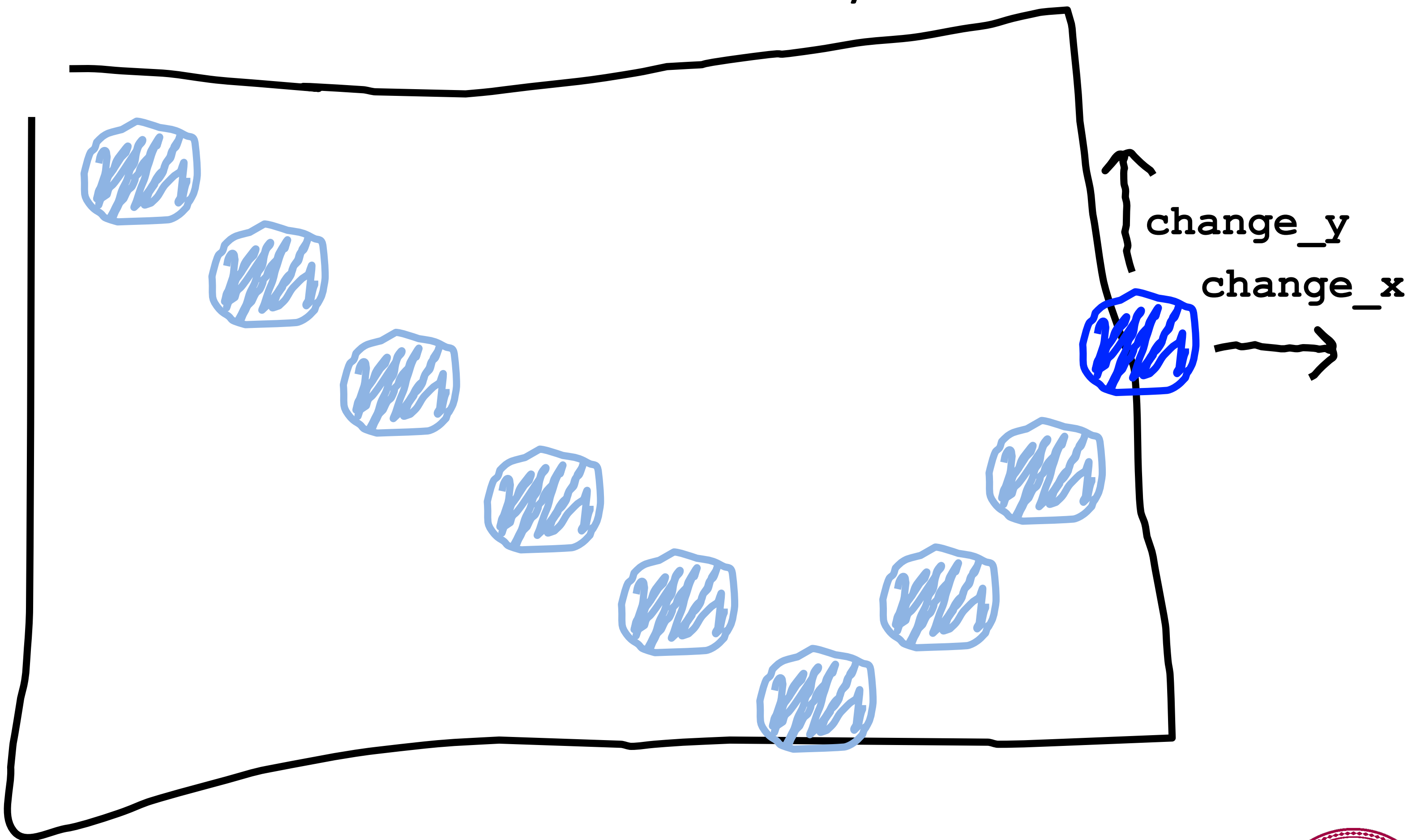
Bouncing Ball

We want this!



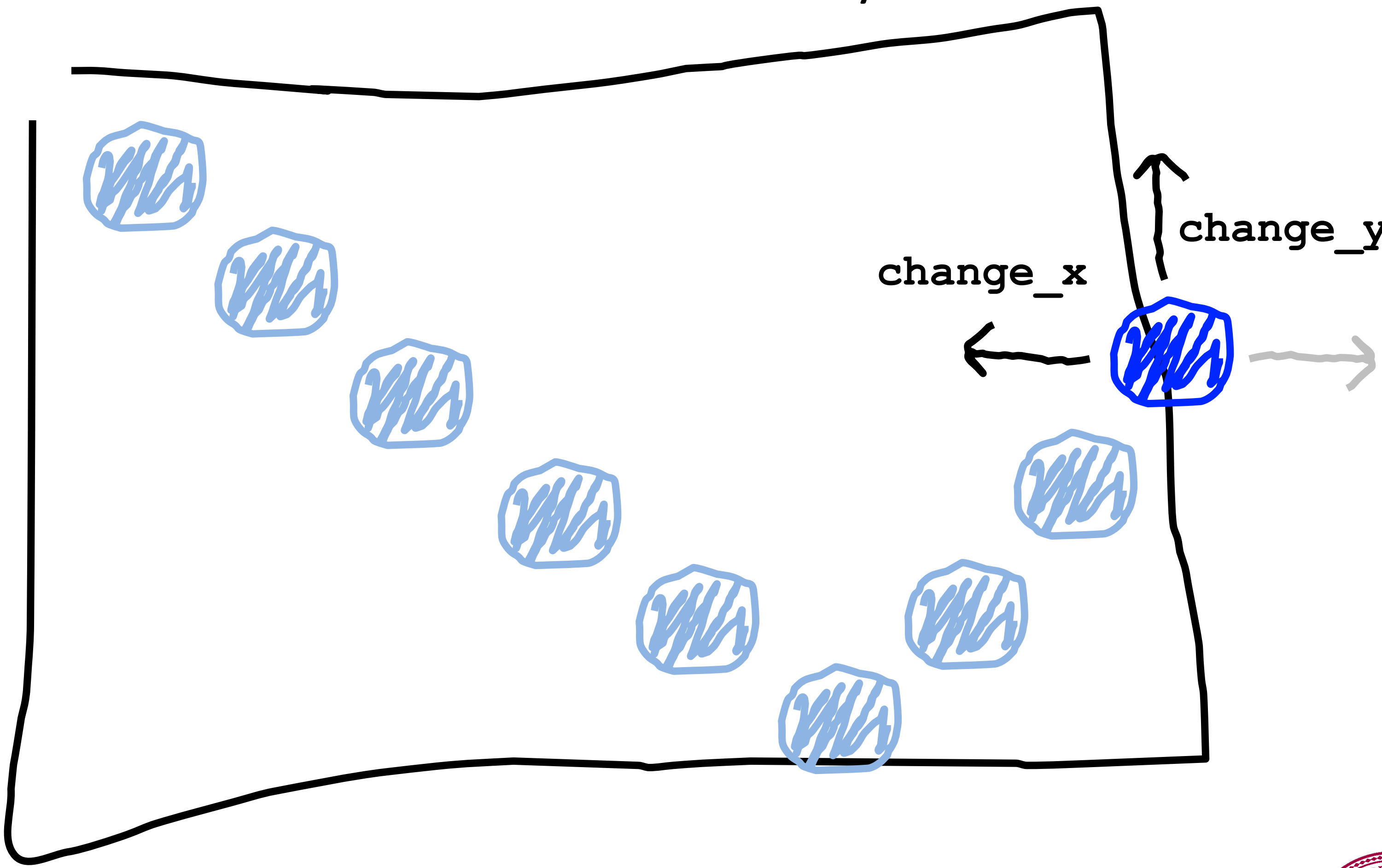
Bouncing Ball

This was our old velocity



Bouncing Ball

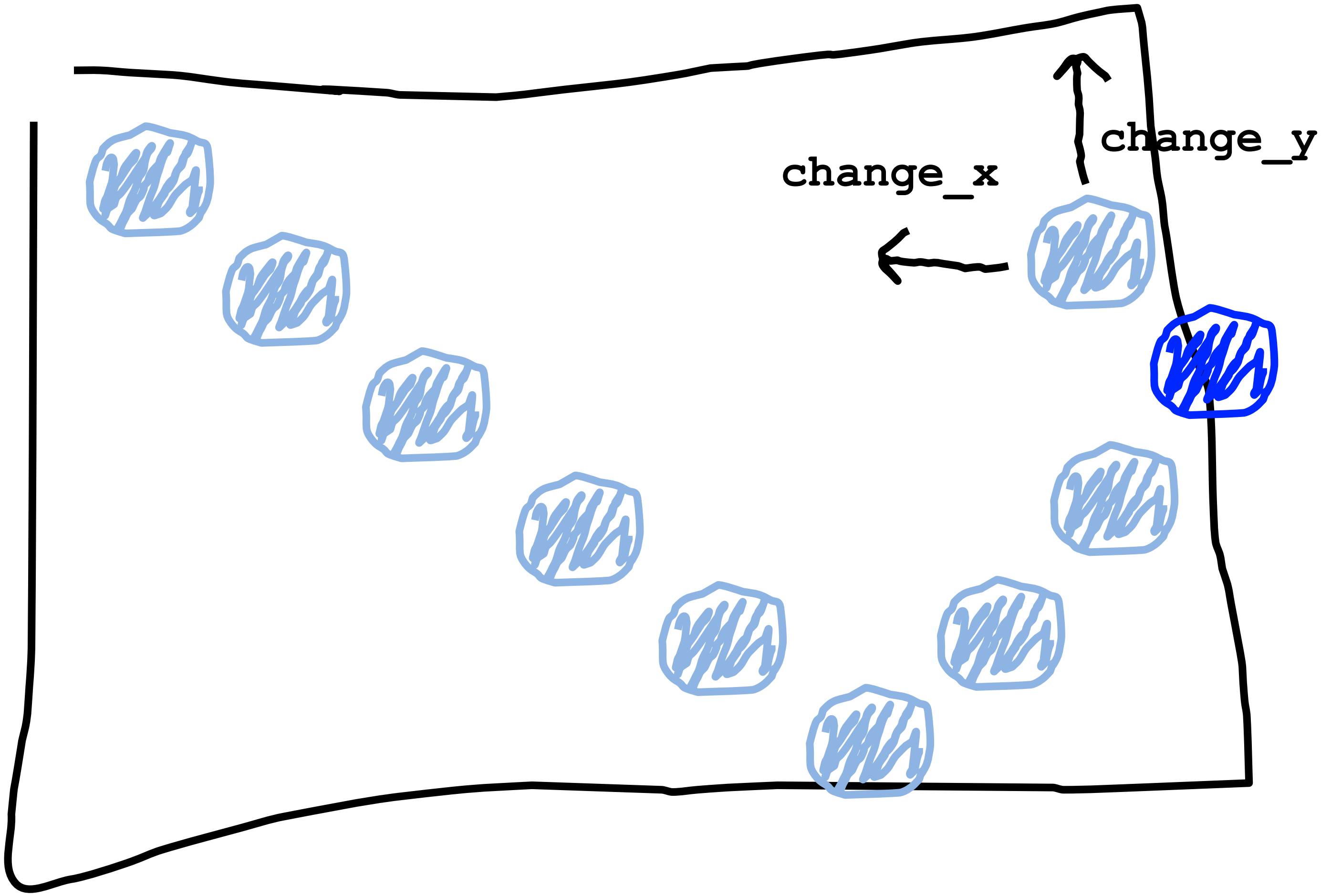
This is our new velocity



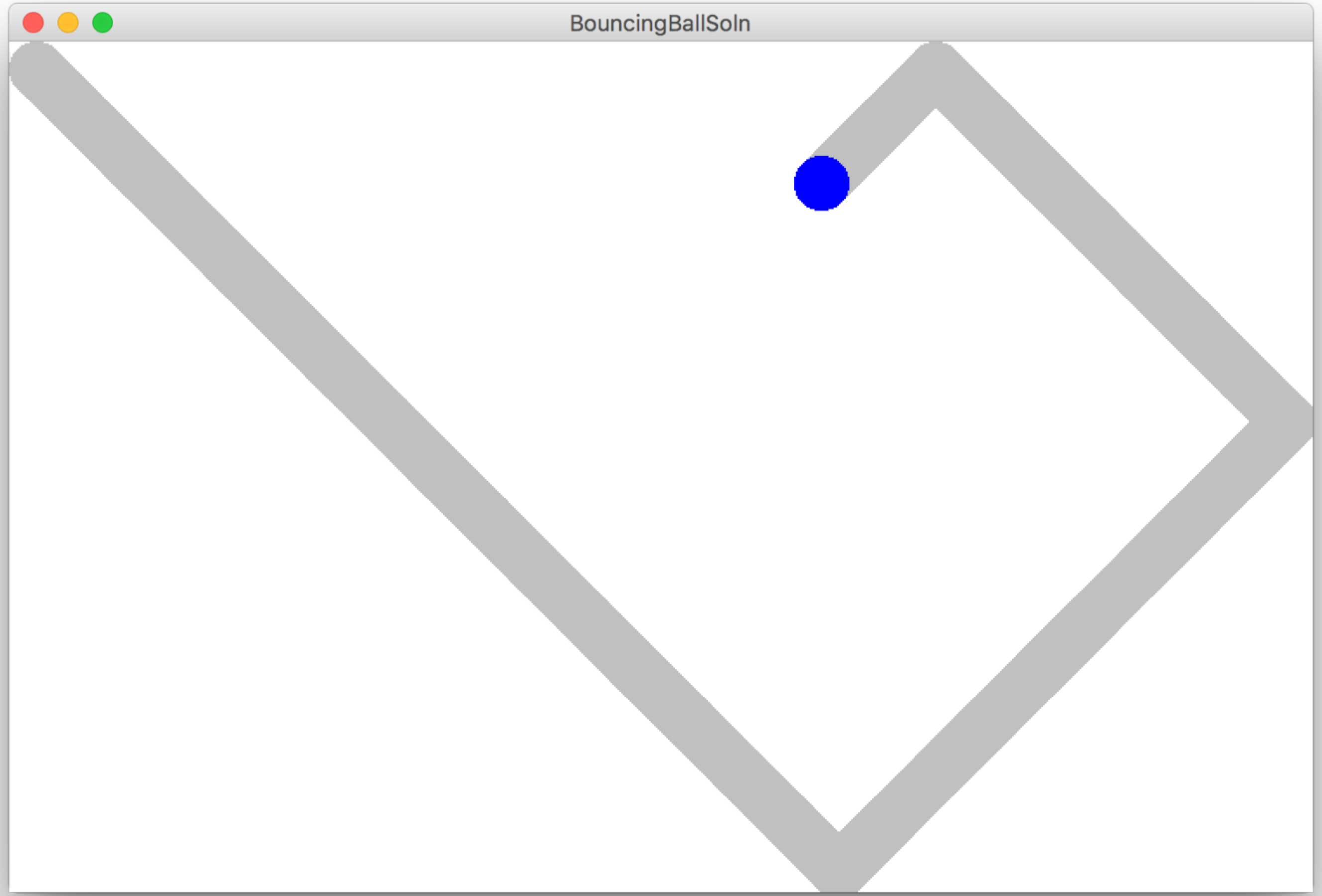
When reflecting horizontally: **`change_x = -change_x`**

Bouncing Ball

Tenth heartbeat



Bouncing Ball



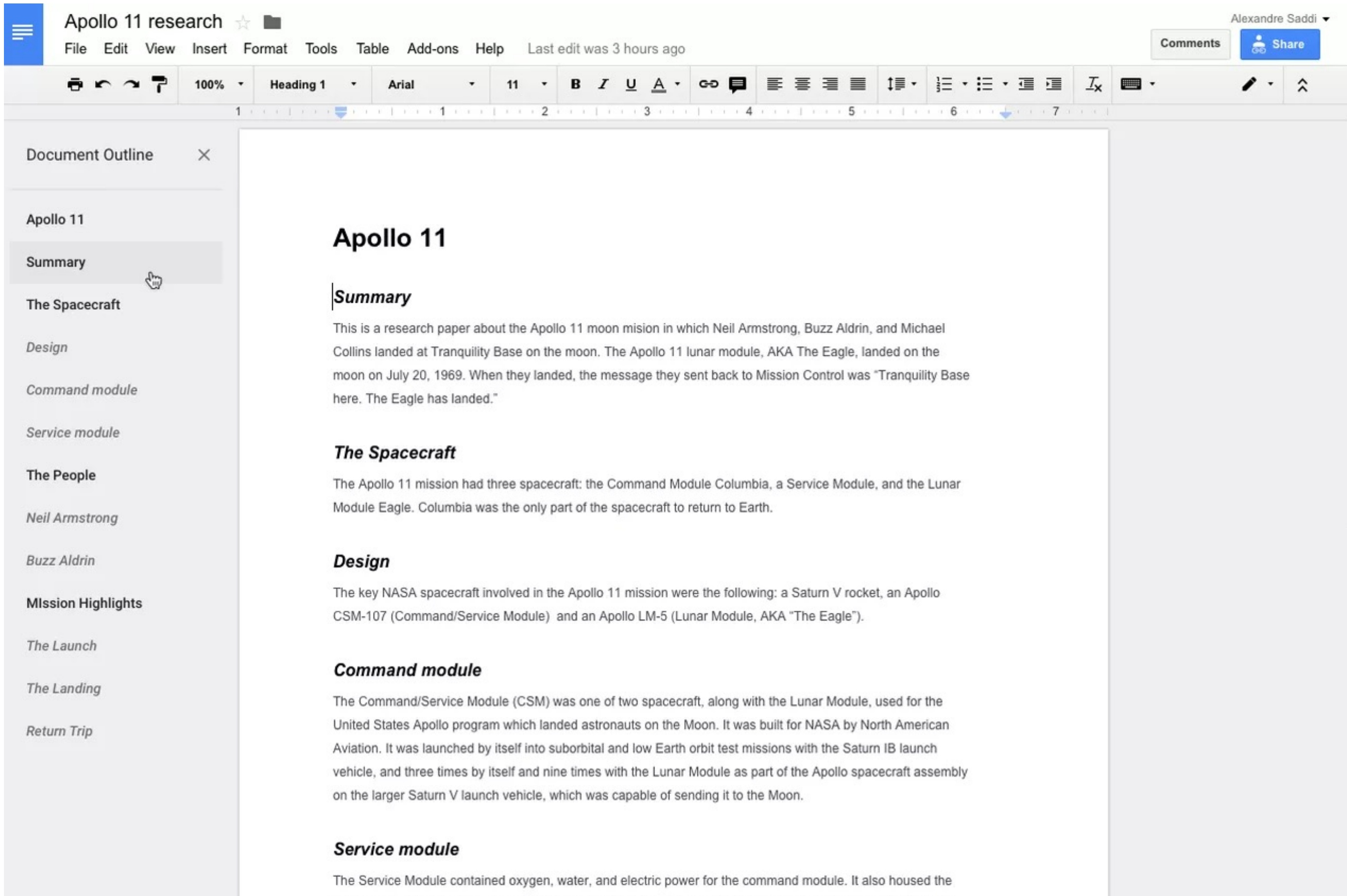
Hold up!

```
def make_ball(canvas):
```

If you get a copy when you pass a parameter. Does this copy the canvas??!!

Some Python variables (like Canvas) are stored using a reference. Which is like a ***URL****. The URL gets copied when you pass the variable.*

How do you share google docs?



https://docs.google.com/document/d/1eBtnEilI3KHefFS-kSAOpXqeSXpbfTTMlmOgj6I9dvk/

```python
def main():
    canvas = Canvas(…)
    make_ball(canvas)

def make_ball(canvas):
    canvas.create_oval( … , 'blue')
```

stack

heap

main

```
def main():
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    make_ball(canvas)

def make_ball(canvas):
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```

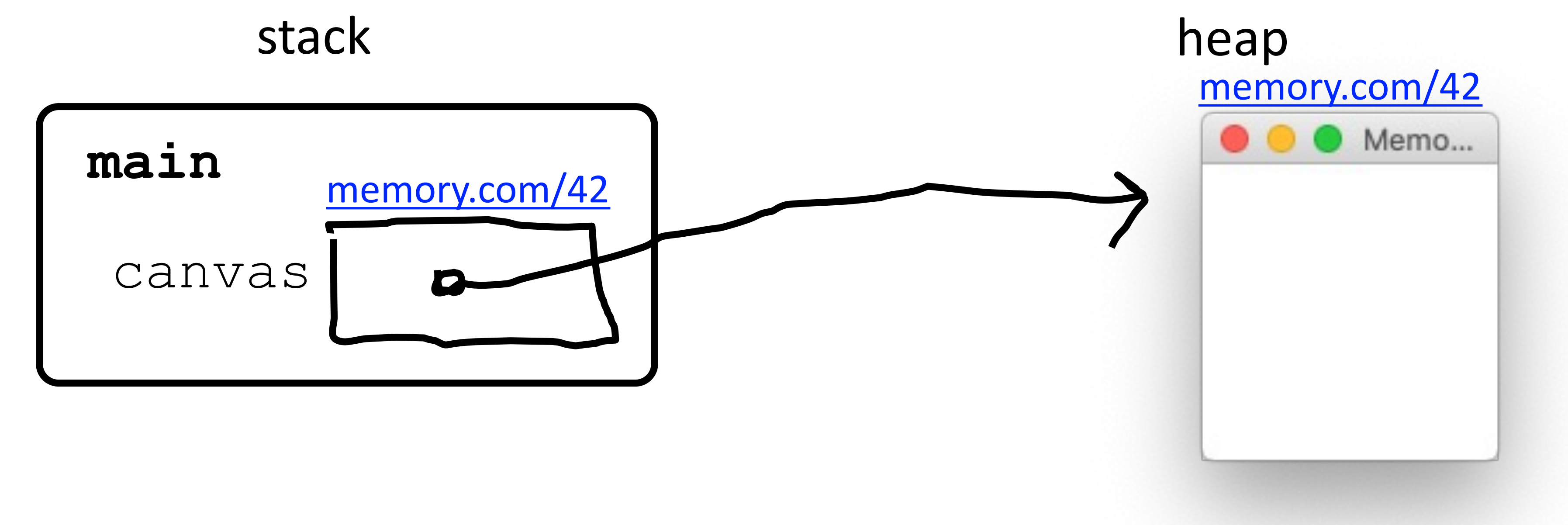

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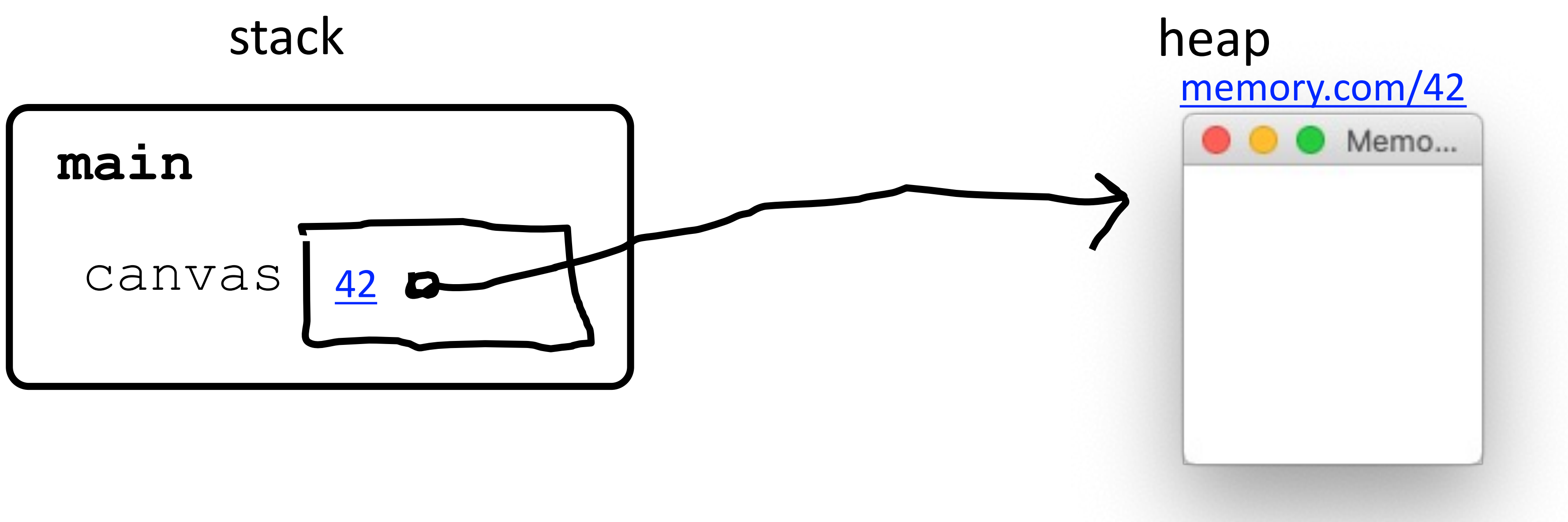

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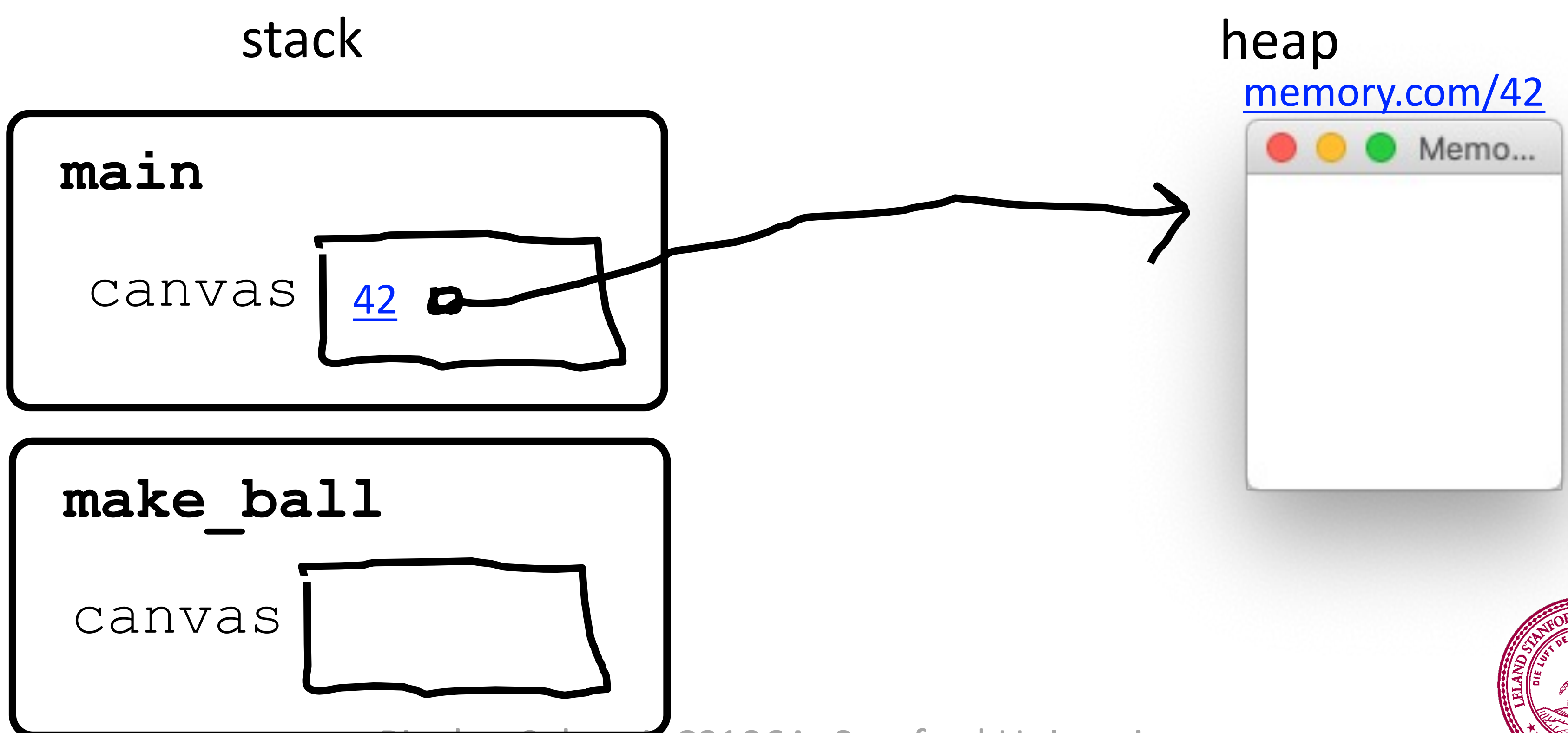

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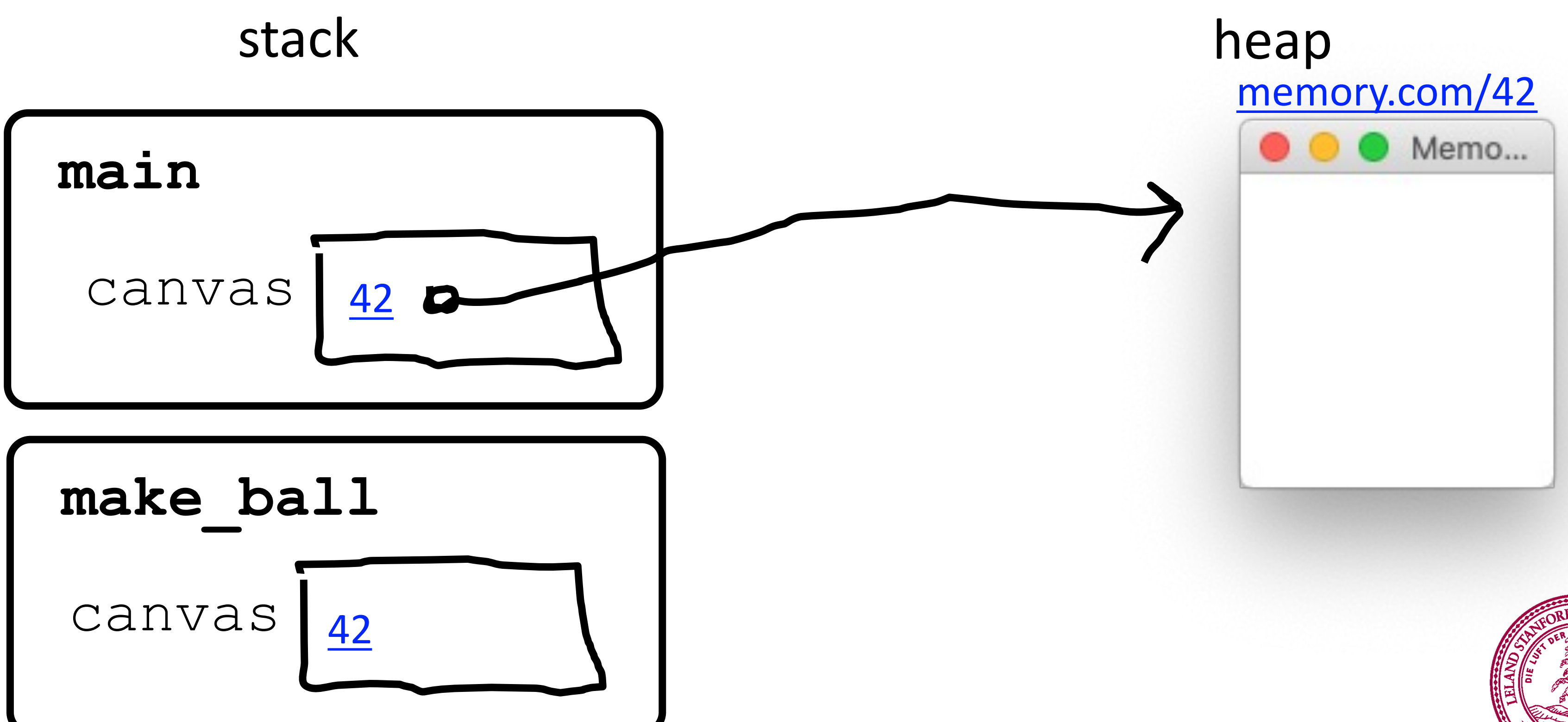

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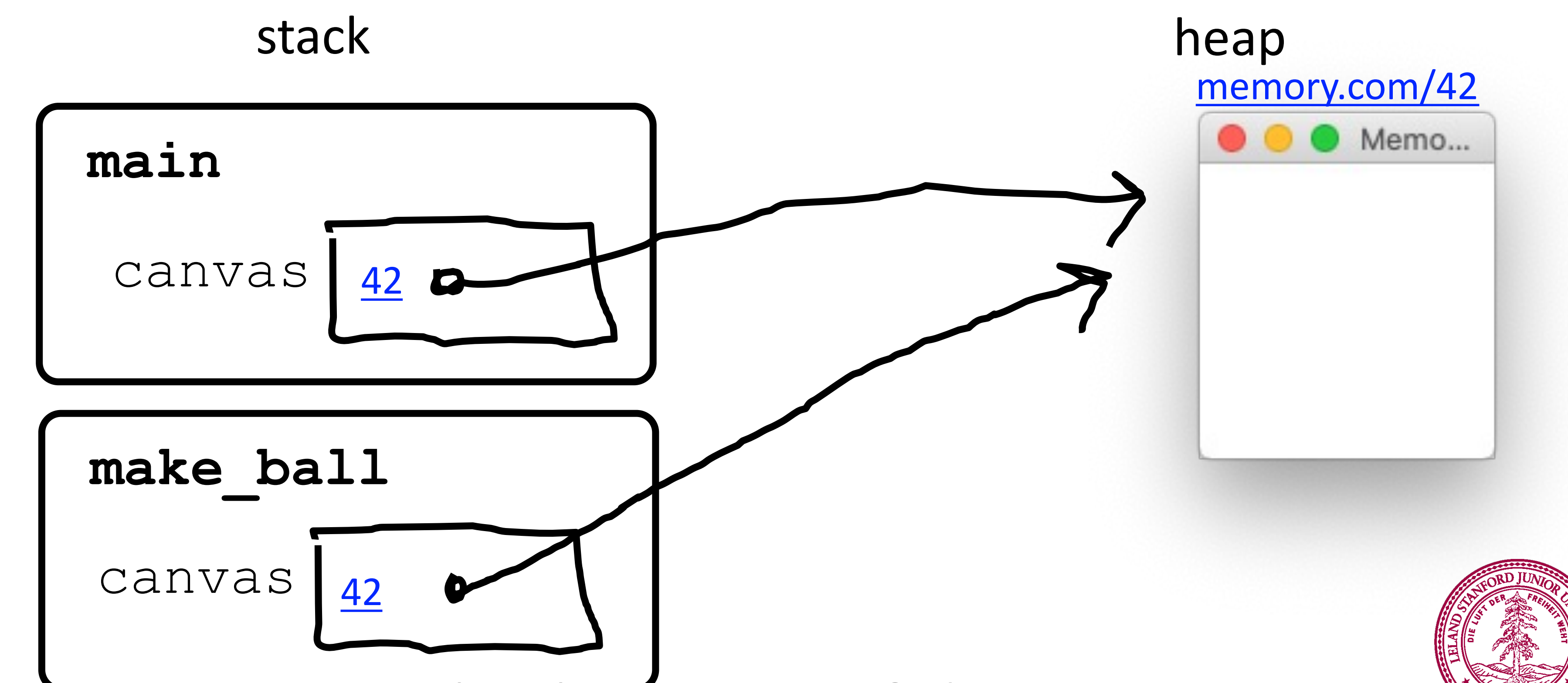

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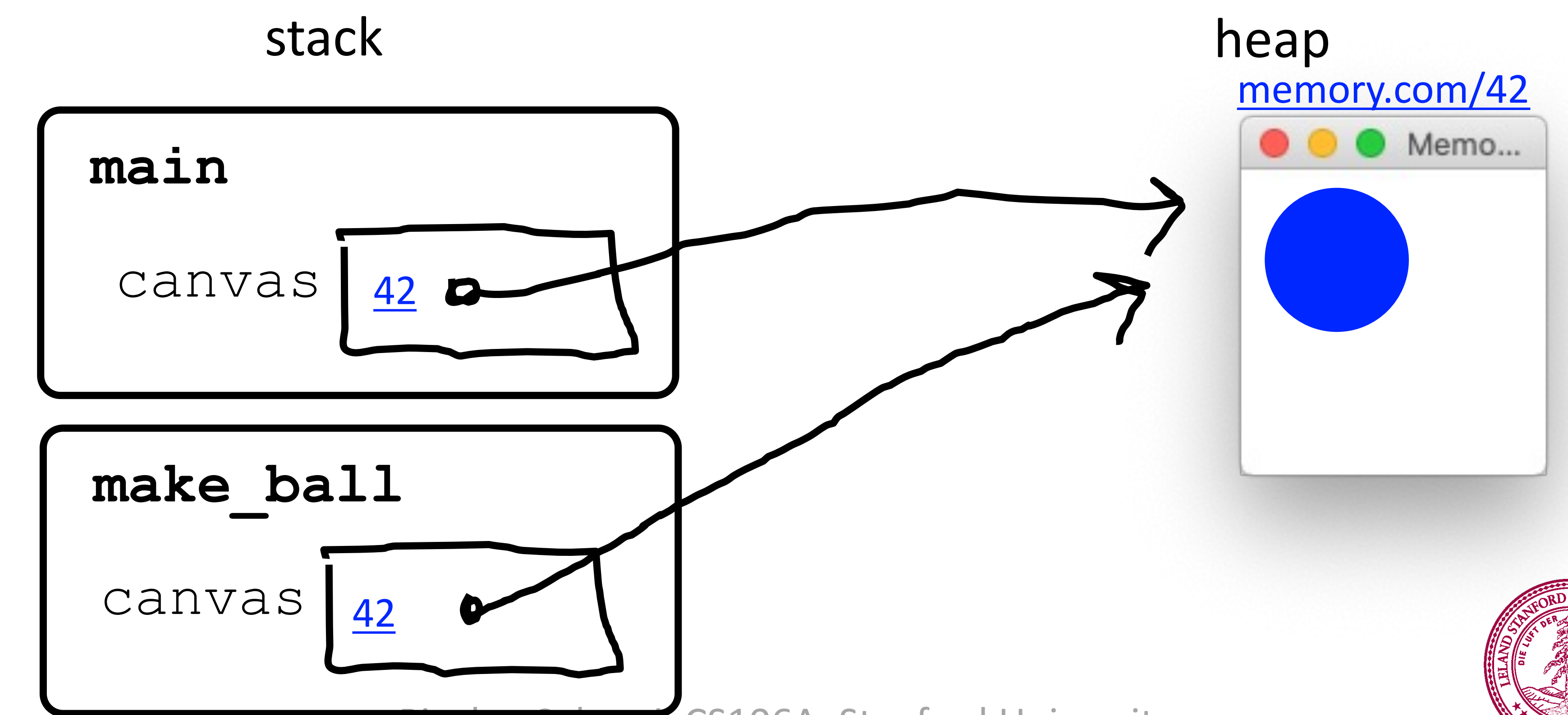

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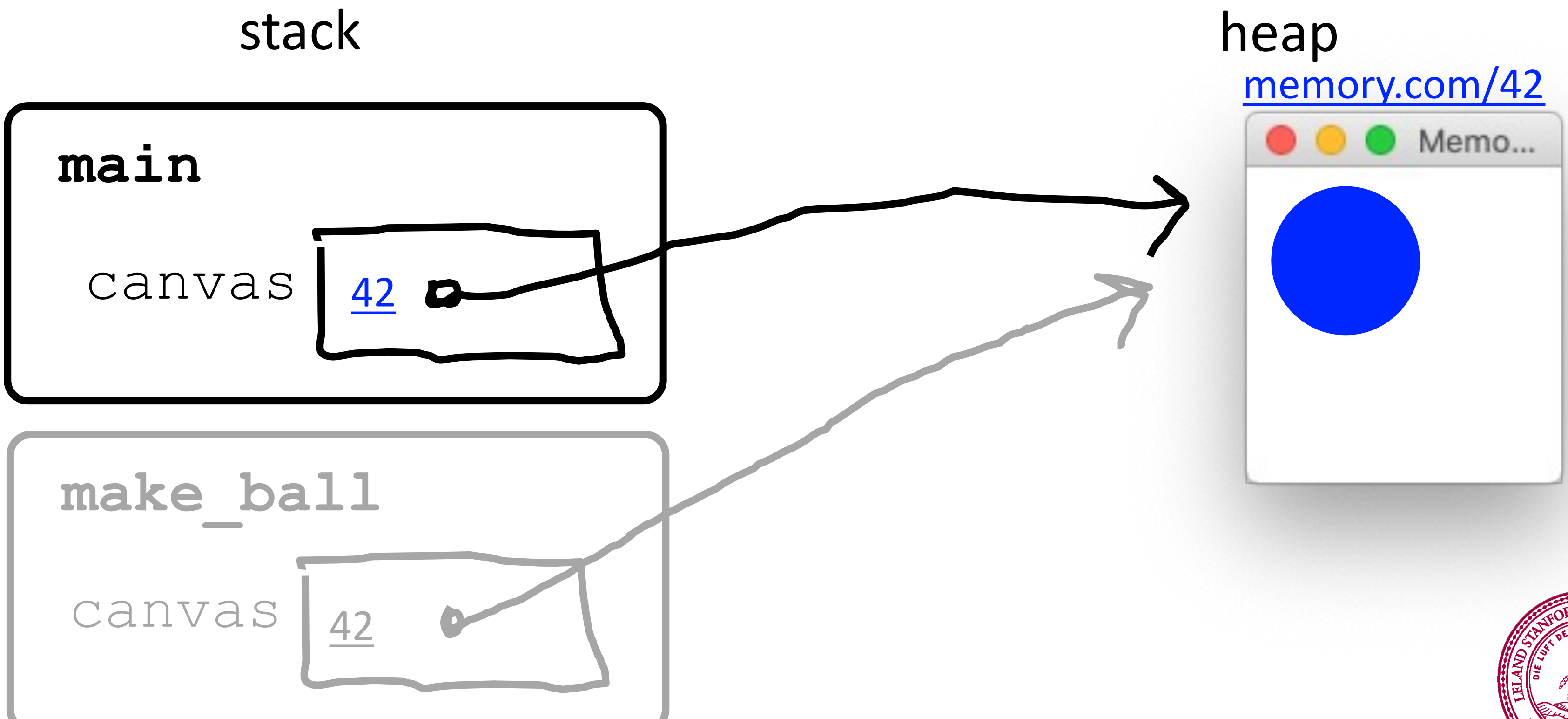

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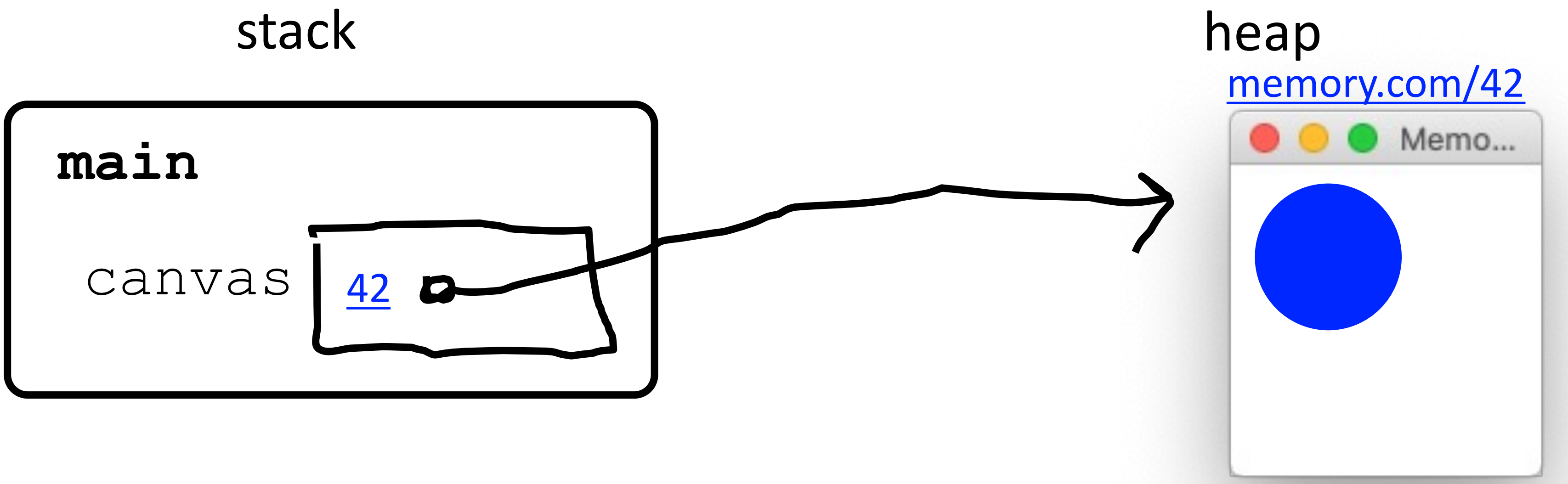

When passing variables, some act just like you are passing a URL.

That allows functions to modify the variable

Special Graphics Functions

```
# get the x and y location of the mouse
mouse_x = canvas.get_mouse_x()
mouse_y = canvas.get_mouse_y()

# move shape to some new coordinates
canvas.moveto(shape, new_x, new_y)

# move shape by a given change_x and change_y
canvas.move(shape, change_x, change_y)

# get the coordinates of a shape
top_y = canvas.get_top_y(shape)
left_x = canvas.get_left_x(shape)

# return a list of elements in a rectangle area
results = canvas.find_overlapping(left_x, top_y, right_x, bottom_y)
```

Let's track the mouse!
(Fun for things like games)